Journal of Public Economics 248 (2025) 105436

Contents lists available at ScienceDirect

Journal of Public Economics

journal homepage: www.elsevier.com/locate/pubec

To burn a slum: Urban land conflicts and the use of arson against *Favelas*

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HIGHLIGHTS

- Fires are disproportionately more likely to occur in slums located on the most valuable lands in São Paulo.
- Relationship between fires and land value exists only in slums occupying private lands, indicating the occurrence of economically motivated arson.
- Probability of fire increases non-linearly, which is consistent with theory, and is not explained by differences in infrastructure or density.
- Results are supported by both a cross-sectional analysis and a Difference-in-Differences exploring the impact of an urban renovation program.

ARTICLE INFO

Keywords:
Urban land conflict
Slums
Arson
Violence
Land value
Property rights

ABSTRACT

This paper investigates the understudied phenomenon of urban land conflicts in contexts with weak enforcement of property rights. I examine, both theoretically and empirically, the use of arson as a violent tool to force slum removal from high-value land in cities. Leveraging fine-grained geocoded data, I employ panel regression and Difference-in-Differences analyses to demonstrate that the probability of slum fires dramatically increases with rising land prices. This effect is nonlinear and driven exclusively by slums situated on private lands, highlighting the role of high-powered incentives behind arson. These results illustrate how urban land conflicts can have different outcomes than their rural counterparts.

1. Introduction

A growing body of evidence in Economics has shown that weak enforcement of land property rights can lead to violence (Alston et al., 2000; Fetzer and Marden, 2017). When formal institutions fail to provide legal solutions, squatters and landowners often fight to gain control of contested land. While extensive attention has been devoted to understanding this phenomenon in rural areas, little is known regarding the emergence of violent land conflicts in urban settings (Lombard and Rakodi, 2016). Such conflicts are likely to arise in many cities, where landowners and slum residents compete for valuable land, but local governments often lack the capacity or motivation to resolve these disputes (Brueckner and Selod, 2009; Jimenez, 1985; Lanjouw and Levy, 2002; Holston, 1991; Besley and Persson, 2010). This is expected even when slum-occupied land is private and property rights should be better enforced.[2]

This paper provides novel evidence on whether and how urban land conflicts become violent. I investigate both theoretically and empirically the use of arson as one potential strategy in urban land conflicts—a possibility suggested in many settings. In the US, cases of arson-driven displacement of tenants in Hoboken (NJ), Boston (MA), and other urban centers were likely associated with gentrification during the 1980s

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[1] I thank Sergio Firpo, Fernando Ferreira, Rodrigo Soares, Eduardo Faingold, Mariaflavia Harari, Maisy Wong, Bruno Barsanetti, Paulo Furquim, and Leila Pereira, as well as all seminar and conference participants at Insper, Wharton, the 2018 Ronald Coase Institute Workshop, the 2019 Urban Economics Association Summer School at LSE, the 44th EBE-SBE, the Paris School of Economics Workshop on Neighborhoods and Local Interactions, and the 2023 LACEA-LAMES for their comments and contributions. I also thank the editors and referees for all their insights and suggestions. Finally, I thank the São Paulo State Research Foundation (FAPESP) for their financial support (scholarship grants 2018/14183-3, 2019/02389-9, 2023/11134-0).

[2] Contestability of private land can arise, for example, when there is room for adverse possession, i.e., when legislation allows for squatters to become legal owners of the squatted land if they meet some criteria (Handzic, 2010; Baker et al., 2001).

https://doi.org/10.1016/j.jpubeco.2025.105436
Received 21 August 2024; Received in revised form 18 April 2025; Accepted 12 June 2025
Available online 25 June 2025
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(Gottlieb, 2019; Brady, 1983).[3] The wave of what became known as *arson-for-profit* was even discussed in legislative hearings in the US Congress (U.S. Congress, 1980).

In developing countries, where the enforcement of land property rights tends to be weak, arson is allegedly used to destroy slums in premium neighborhoods, force residents out, and clear the land for sale or development (Ockey, 1997; Rahman, 2001; D'Andrea, 2012; Barcelos and Viana, 2017; Malhotra, 2018). For urban landowners, using this strategy could be more advantageous than engaging in overt armed confrontation, which is typically the case in rural land conflicts. Because fires in slums spread fast, it is hard to determine their true causes, let alone find potential arsonists (Braga and Landim, 2008; Walls et al., 2017). Therefore, landowners could use arson to enforce their claim over contested land while still evading the high-monitoring environment of dense urban areas.

To examine the use of arson, I study how the incentives for engaging in violent land conflict vary according to prices and legal ownership (private or public) of slum-occupied land. On the one hand, higher land value drives higher willingness to remove slums (Jimenez, 1985; Turnbull, 2008; Brueckner and Selod, 2009; Brueckner et al., 2019). I refine this idea with an optimal stopping model, suggesting a non-linear relationship between land value and removal. On the other hand, ownership matters: private landowners should have stronger incentives to remove slums than public entities. While private owners make an individual decision, public owners face a collective action problem that may reduce their incentives to intervene (Olson, 1971).

To test these implications, I construct a unique database spanning from 2001 to 2016, incorporating geocoded information on slums, fires, and land prices in São Paulo, one of the world's largest cities (United Nations, 2018). With a cross-sectional analysis and a Difference-in-Differences design, I explore variation in both land value and ownership across slums and over time, examining how these factors affect the prevalence of arson against slums.

São Paulo provides an interesting setting to study arson and urban land conflicts for three main reasons. First, as in other cities in the developing world, São Paulo has had many cases of fires reportedly intended to remove slums (Mahmud, 2016; Rahman, 2001; Malhotra, 2018; Ockey, 1997).[4] These events gained such proportion that an inquiry commission was instituted by the city council. Although hard proof of wrongdoings is scarce, there is suggestive evidence that a large share of slum fires in the city are caused by arson (Bruno, 2010).[5] Second, the city has slums on both private and public lands spread across several neighborhoods with varying land prices. Finally, there is rich data available for a long period of time—a rarity when studying slums.

The main hypotheses to be tested are derived from a dynamic framework inspired by classical models of real estate development (Capozza and Sick, 1994). In my context, I show that landowners decide whether to set fire to slums based on whether land value exceeds a critical threshold. When it is sufficiently high, it becomes optimal for the landowner to burn the slum and proceed with development, rather than waiting for legal institutions to remove the occupants. Conversely, when land value is below this threshold, delaying action and relying on legal mechanisms is preferred. This model predicts that the relationship between arson and land value is not only positive but also non-linear, featuring a discontinuous jump in the likelihood of arson once land value crosses the critical threshold.

Such behavior should be mostly driven by private landowners because they can retain all the value from removing the slum. If the land is public, there are some detracting factors. First, removing slums from government land could be interpreted as a publicly provided urban amenity for neighboring property owners, since it may positively affect house prices (Shah, 2014). In such case, there is a classical coordination problem resulting in under-provision of removals. Second, there can be reputational costs for incumbent authorities, since people tend to blame them for not preventing fires. Finally, governments are often required to provide aid to displaced victims of fires.

I design two empirical exercises to investigate these theoretical results. In both cases, the main challenge arises from not observing arson directly, but rather slum fires in general. This type of measurement error can lead to omitted variable bias if slums with higher land value are also closer to jobs and amenities, drawing more residents and being subject to more domestic accidents involving fire, for instance.

First, I explore cross-sectional variation in slum land value to test whether the probability of arson increases non-linearly. I run a linear probability model to compare the yearly incidence of fire for slums in different quantiles of land value and across private and public lands. I rely on multiple features to mitigate potential omitted variable bias: (i) several controls for observable characteristics related to land value and fire hazard, such as slum density and infrastructure; (ii) district-year fixed effects to compare slums in similar locations; (iii) the fact that slums were built decades prior to current land price dynamics, which should limit sorting; (iv) the interaction of land value with private/public land ownership is exogenous to fires.

I find that the probability of fire increases discontinuously for slums in the highest quantiles of land value. However, the effect only exists for slums on private land. There is no evidence of strategic arson on public land.

I then run a Difference-in-Differences to provide more robustness to these findings. The estimated effect is more localized than in the cross-sectional analysis, but there is a clearer causal interpretation. I leverage a plausibly exogenous shock to land values caused by a large urban renovation program in São Paulo. Starting in 2004, the municipal government designated a specific area to receive large investments in urban infrastructure. To raise funds, it auctioned permits for developers to build above zoning restrictions in that area. The perspective of better urban amenities coupled with the possibility of constructing taller buildings attracted development, causing demand for land to increase, as well as land value.

I use the locations of slums relative to the intervention area to categorize their exposure to the shock. I then compare the probability of fire across more and less exposed slums before and after the intervention. I also compare outcomes for slums on private and public lands, expecting the effect to be driven mainly by the former.

Again, I find effects consistent with strategic arson. The probability of fire increases substantially for slums within the intervention area—but only when they are located on private land. This result is robust to the inclusion of various control variables, slum and year fixed effects, and alternative restrictions on the sample period and size.

Besides pointing to the same conclusion, results from the Difference-in-Differences and the cross-sectional approaches are consistent with each other. The shock caused by the intervention to exposed slums is sufficient to move them across the theoretical land value threshold that makes strategic arson profitable, as estimated in the cross-sectional model.

Finding evidence of strategic arson to force slum removal contributes to the growing literature about land conflict and weak enforcement of property rights (Alston et al., 2000, 2012; Fetzer and Marden, 2017). I document that urban land conflicts can also become violent, but the outcome is different from what we observe in rural or remote areas of developing countries.

Additionally, I contribute to a fundamentally theoretical literature about slum removal. Using fine-grained information on slums, this

[3] In Hoboken, for instance, rent control in a rapidly gentrifying area drove strong incentives for landlords to displace tenants, renovate units, and rent at a higher price afterwards.

[4] For a more complete documentation of episodes covered in media articles, please refer to the Appendix.

[5] Bruno (2010) shows that 30 % of slum fires in São Paulo in the beginning of the 2000's were caused by arson, although there is no data on the motivation behind these cases.

paper provides a within-city empirical test of the relationship between land value and attempts of eviction (Jimenez, 1985; Turnbull, 2008; Brueckner and Selod, 2009; Shah, 2014; Brueckner et al., 2019). Moreover, I show both theoretically and empirically that such relationship is likely non-linear due to the dynamic nature of real estate development.

Finally, this paper suggests that slum residents face extra—and potentially invisible—costs when the formal city expands towards their neighborhoods. This adds to the conclusions in Henderson et al. (2021), who develop a model to study the transition of neighborhoods from slum to non-slum status. Although we both use an optimal stopping framework, Henderson et al. (2021) are more interested in explaining city growth and how this interacts with informal property rights in slums. I focus on land conflicts that arise before the transition can happen and on how they may become violent.

The remainder of the paper is organized as follows. Section 2 describes institutional factors behind strategic arson in slums and lays out a brief conceptual framework to guide the empirical analysis. Section 3 presents the data. Sections 4 and 5 show the main results of the paper using cross-sectional and Difference-in-Differences analyses, respectively. Section 6 concludes.

2. Institutional background and conceptual framework

2.1. Land property rights and slum removal

Land conflict is common in Brazil and often stems from weak enforcement or contestability of property rights (Alston et al., 2000, 2012; Fetzer and Marden, 2017; Chiavari et al., 2021). This issue arises not only from the state's limited capacity to resolve disputes but also from convoluted regulations allowing for adverse possession.[6] Although originally designed to support land reform, adverse possession may also lead to judicial uncertainty and promote rather than prevent disputes (Holston, 1991; Gonçalves, 2009).

Lack of land property rights is widespread in Brazilian cities, where slums are built on lands legally owned by other agents, either private or public.[7] In 2000, 31 % of the country's urban population lived in these informal neighborhoods (UN-Habitat, 2020). In São Paulo, the largest city in the Southern Hemisphere, 10 % of the population lived in slums in 2000 (Schneider et al., 2022; Pasternak, 2006). Most of these slums formed in a context of rapid rural-to-urban migration and restrictive zoning laws around the 1980s. Unable to access formal housing, migrants settled in peripheral areas, building their own homes on land they did not legally own (Rolnik, 1995).

Since informal property rights can be contested by legal landowners, slum residents are constantly vulnerable to removal (eviction), among other economic consequences (Field, 2007; Galiani and Schargrodsky, 2010; Marx et al., 2013). Removals are typically court-mandated and managed by local authorities. These events involve demolishing houses and displacing residents, with no guarantee of government assistance. Data from Observatório das Remoções (2022) show that between 2017 and 2022, more than half of all slums in São Paulo either had been cleared or faced some risk of removal.[8]

Still, slum removal is uncertain, even when the legal landowner has clear property rights. On the one hand, residents might resist eviction, especially if adverse possession is possible or if the removal could hurt mayors' electoral outcomes. On the other hand, court decisions can take many years to conclude.[9] According to Observatório das Remoções (2022), only 15 % of slums under risk of removal have actually been removed either completely or partially between 2017 and 2022.

2.2. Real estate value and slum land ownership

Land value is an important factor governing the pressure for slum removal. As real estate prices increase, alternative uses for slum-occupied land become more profitable. For instance, the legal landowner could build an apartment complex and rent the units on the land instead of having slum residents living there for free. Such changes in real estate value around slums could be related to gentrification of central areas or even city expansion towards some peripheral neighborhoods (Brueckner and Selod, 2009; Gechter and Tsivanidis, 2018; Henderson et al., 2021).

The influence of real estate value on slum removal likely depends on land ownership, with stronger incentives on private land. Data from Observatório das Remoções (2022) show, for instance, that slums on private land are twice as likely to be removed as those on public land.

Private landowners face an individual decision problem: they must weigh the potential profit from developing the plot after slum eviction against the cost of pursuing the removal (Jimenez, 1985; Hoy and Jimenez, 1991; Brueckner and Selod, 2009; Selod and Tobin, 2018). In contrast, slum removal on public land poses a collective action problem. While eviction may be viewed as an urban amenity that enhances surrounding property values, the benefits are diffuse among many agents (Shah, 2014). As a result, the classical outcome is under-provision of this public service (slum removal). If some individuals value the removal less than others, there is insufficient pressure for local authorities to act.

The government may also have a direct interest in removing slums, but the gains are unclear. Revenue from selling the land may not benefit mayors or their allies directly. Additionally, there are political risks for incumbents, since evicted slum residents often earn sympathy from the unaffected population. Finally, there can be financial costs, as local regulations sometimes require the government to provide assistance to displaced slum residents.

2.3. Urban land conflicts and strategic arson

Conflict may arise when landowners seek slum removal—potentially motivated by rising land values—but face significant obstacles. Since slums are contested territories, disputes can escalate into violence if authorities fail to promote a legal solution (Lombard and Rakodi, 2016). This has been well documented in rural areas, where landowners and squatters arm themselves and clash over land (Alston et al., 2000; Fetzer and Marden, 2017).

In cities, however, strategic arson might be preferred to overt armed conflict due to the stronger presence of formal institutions and law enforcement. Slum houses are often built with highly flammable materials, allowing fires to spread rapidly and making it difficult to determine their exact origin and cause (Braga and Landim, 2008; Walls et al., 2017). Hence, setting fires to slums may serve as a covert tool to efficiently clear them and make land available for alternative use while leaving little evidence of wrongdoing (D'Andrea, 2012; Ockey, 1997; Rahman, 2001). A similar reasoning was arguably behind the arson-for-profit events in the US in the 1980s, where landlords would set fires to displace tenants in gentrifying areas (Gottlieb, 2019).

Discussion about strategic arson in slums is neither new nor specific to Brazil. In New Delhi, a fire devastated a slum while courts were still deliberating on the residents' eviction (Malhotra, 2018). In Bangkok, Ockey (1997) documents a case where residents caught an

[6] In Brazil, adverse possession is regulated by both the Constitution and additional federal laws (Civil, 1988; Brasil, 1981, 2002).

[7] Legal ownership over the land is also known as *de jure* ownership. This is contrasted with *de facto* ownership, in which the owner may informally possess the land. This would be the case of slum residents. A more detailed explanation about *de jure* and *de facto* ownership and its relationship with land conflicts is provided in Alston et al. (2012).

[8] Observatório das Remoções (2022) collects both self-reported and official information to try and map slum removals in the city.

[9] Among repossession lawsuits for formal houses in the city of São Paulo, which are supposed to be simpler than those involving slums, 25 % take more than 2 years to be ruled and some even last up to 9 years.

arsonist attempting to ignite a fire in a slum under threat of eviction. Similarly, Rahman (2001) reports at least five instances of arson used to displace squatters in Dhaka in the 1990s.

In Brazil, a large slum in Rio de Janeiro was destroyed by a suspicious fire in 1969. After resisting eviction tied to an urban renewal project, residents were ultimately displaced following the blaze (Brum, 2011). More recently, news reports suggest a similar pattern in São Paulo, where fires in high-value neighborhoods appear to be driven by economic incentives.[10]

There is also a strong correlation between fires and slum removals. Among a sample of 86 slums removed between 2001 and 2016 in São Paulo, 60 % experienced a fire during the same period.[11] Moreover, the probability of slum removal in a given year is seven times higher if the slum experienced a fire that year or in the previous one. Hard evidence on strategic arson, however, is still scarce. For São Paulo, Bruno (2010) estimates that 30 % of slum fires in the early 2000s were deliberately set, but the motivation behind them is unclear.

2.4. Conceptual framework

This section provides a brief theoretical framework to motivate the empirical analysis. The main implication is that the probability of arson is positively correlated with land value, but in a non-linear fashion. It should be higher than zero only for slums located in neighborhoods where land value is above a given threshold. The model builds on Henderson et al. (2021), Capozza and Li (1994), and Capozza and Sick (1994) and its full derivation is presented in Appendix B.

In this context, landowners face a dynamic decision: whether to wait for a court-mandated slum removal or to force it through arson. Courts may rule in favor of eviction, but there is uncertainty regarding the outcome and the duration of trials. Alternatively, burning the slum irreversibly clears the land for alternative use. By committing arson, however, landowners risk being arrested.

Real estate value governs the dynamics of this decision. As prices increase, so does the expected profit of clearing the slum. This means that waiting for a court decision entails an opportunity cost that is potentially growing over time. At sufficiently high prices, the cost of waiting another period can outweigh the risk of committing arson to clear the slum immediately.

Consequently, the probability of arson evolves non-linearly with land value. There is a threshold above which arson becomes profitable, also influenced by institutional factors such as the cost of arson, the likelihood of conviction, the probability of a favorable court ruling, and other legal costs.[12]

3. Data and descriptive analysis

To test the relationship between land value and the probability of arson, I rely on information about the location of slums, the number of fires in each of them, and some measures of value for the land they occupy. The following subsections provide details about these different pieces of information.

[10] Appendix A provides news reports on suspicious slum fires that may have been economically motivated.

[11] Observatório das Remoções (2022) contains self-reported observations. I used Google Earth images to verify, for a subsample, whether slum buildings were visibly destroyed.

[12] As detailed in Appendix B, arson only becomes an option if the efficiency of courts and the probability of getting caught are sufficiently low. This aligns with the reality in São Paulo, where eviction lawsuits can be lengthy, and arson convictions are rare. Between 2011 and 2016, comparing the number of arson cases under investigation or in court with total fires in the city yields a ratio of less than 10 %. Conversely, Bruno (2010) suggests that 30 % of all fires in the city were caused by arson in the early 2000s.

Table 1
Descriptive statistics of slums and neighborhood in 2000.

	Private land		Public land	
	Mean	S.D.	Mean	S.D.
Census				
Share piped water	0.96	0.11	0.96	0.12
Share waste collection	0.96	0.10	0.97	0.07
Share low income	0.42	0.12	0.41	0.11
Share no income	0.15	0.09	0.16	0.09
Residents per household	3.78	0.33	3.79	0.28
Households per hectare	1332.05	2483.36	1496.94	3128.55
Municipal government				
Area in hectares	1.68	4.74	1.32	4.47
Share landslide risk	0.20	0.30	0.23	0.32
Year occupation	1974.11	11.58	1977.85	9.19
Number of slums	453		1552	

Notes: 'Private Land' denotes slums on land that is either partially or totally owned by private agent. 'Public Land' denotes slums on land that is either owned by the government or whose ownership is unknown.

3.1. Slums

The municipal government of São Paulo provides geocoded, cross-sectional data for more than 2000 slums in the city collected between 2001 and 2003. Slums are defined as precarious, spontaneous, and self-constructed settlements with no street planning, no land property rights, poor infrastructure, and low-income families (Prefeitura de São Paulo, 2022). Apart from location, the data also document whether legal land ownership is private, public (government), or unknown.

Approximately 23 % of slums are located on land partially or fully owned by private agents. The remainder belongs to various levels of government or has unknown ownership. The latter category accounts for 10 % of all slums and does not necessarily indicate missing data, as unknown land ownership is relatively common in Brazil.[13] Throughout the paper, I treat these lands as public and include them in the analysis to improve precision. This assumption appears reasonable, as private owners should have stronger incentives to assert their property rights, making it less likely that their land is classified as "unknown". Nevertheless, I discuss how this category affects results with robustness checks.

Using a spatial merge, I incorporate data from the 2000 Brazilian Census into the slum polygons, including the number of residents and households, income levels, and access to public services. Table 1 provides descriptive statistics at slum level and is split by land ownership status. Census variables should be interpreted as a blend of slum and neighborhood characteristics, because Census Tracts and slum polygons rarely align perfectly.[14] This helps explain the relatively high reported access to public services, for instance.

Table 1 indicates that slums on private and public lands are quite similar on average. Furthermore, Fig. 1 shows that these two types of slums are not clustered, but rather well distributed across the city. It also reveals that few slums are located close to the city center (represented by black triangle).

Despite this geographical distribution, slums are not confined to low-income regions. Since the 1980s, São Paulo has undergone significant and uneven expansion, with rich households settling in peripheral areas. As a result, slums often coexist alongside high-income neighborhoods, frequently sharing boundaries. Appendix Fig. C.1 illustrates this pattern, showing that nearly one-fifth of slums in São Paulo are surrounded by a higher-than-average share of high-income households. Moreover, Appendix Fig. C.2 suggests that the city is growing towards some slums in peripheral areas.

[13] According to Freitas et al. (2018), ownership cannot be determined for nearly 20 % of the Brazilian territory.

[14] To merge Census data with slums, I assign weights proportional to the area of intersection between slums and Census Tracts.

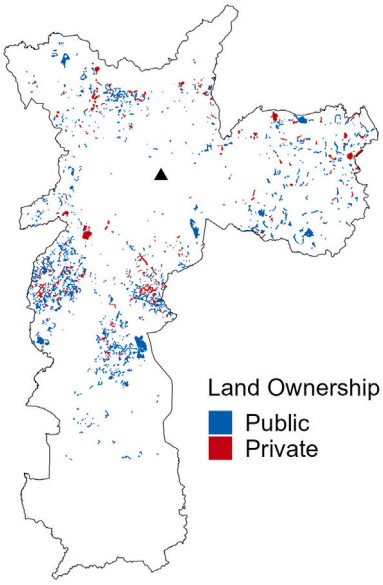


Fig. 1. Spatial distribution of slums in São Paulo in 2001–2003 period and land ownership.
Notes: Figure depicts the municipality of São Paulo and its slums categorized by the legal ownership of slum-occupied land. City center is represented by a black triangle. *Private* includes lands that are entirely or partially owned by private agents. *Public* includes both lands owned by any level of government and those whose ownership is unknown.

3.2. Fires

Data on fires range from 2001 to 2016, comprise the entire state of São Paulo, and were provided by the São Paulo State Fire Department (COBOM-SP). I observe whether the fire happened in a slum or not, as well as a street address. There are no details about size, duration or cause of fires.

Due to changes in recording methodology, the fire department divided the database into three periods: (i) 2001–2006, (ii) 2008–2010, and (iii) 2011–2016. Data for 2007 is entirely missing from the archive. Additionally, address information is less precise for period (ii), as municipality names are absent. Instead, only district names are available, which are not always unique across municipalities in the state. For (iii), exact street address is often missing, but the fire department posted this information on their social media for almost all slum fires during this interval.

As a consequence, geocoding success rates vary across periods. Appendix Fig. C.3 illustrates this, showing that rates are comparable from 2001 to 2010 (around 10 %), but much higher from 2011 to 2016 (average of 42 %). Although inconvenient, this sort of problem is expected when working with data on informal activities, even when authorities are committed to producing high-quality information.

As long as accuracy issues are not correlated with slum land value, however, this poses little concern. To mitigate potential issues, I include year fixed effects in all specifications. I also run robustness tests restricting the sample period in multiple ways.

Table 2 provides the yearly average number and probability of fires in slums. Overall, slums on private land observe more fires than those on public land. However, the difference is not substantial given the standard deviations.

Moreover, frequency and probability are similar across public and private lands because there are few cases of multiple fires in the same slum within a year. The likelihood of observing two or more fires in any given year in the same slum is ten times lower than the average probability.

Table 2
Yearly average number and probability of slum fires, from 2001 to 2016.

	Private land	Public land
Number of fires	0.04	0.03
	(0.28)	(0.20)
Probability of fire	0.03	0.02
	(0.18)	(0.16)

Notes: 'Private Land' denotes slums in lands that are either partially or totally owned by private agent. 'Public Land' denotes slums in lands that are either owned by the government or whose ownership is unknown.

3.3. Slum land value

Measuring land value in slum-occupied areas is challenging due to data limitations. To overcome this, I rely on proxies derived from formal housing market data, sourced from annual property tax records provided by the municipal government. I infer slum land value by aggregating indirect measures of formal land value within a 500-meter radius around each slum.

I rely on two alternative measures, each with its own advantages and limitations: (i) government-defined property prices per square meter for tax purposes, known as Assessed Values; and (ii) Floor-to-Area Ratio (FAR), which is total floor space divided by total plot area. They are available for all formal residential and non-residential buildings in São Paulo during the entire sample period. *A priori*, house prices and FAR should correlate well with land value (Brueckner and Singh, 2020; Wen and Goodman, 2013).

Assessed values are estimated by the government based on past market transactions, construction quality, and location parameters.[15] However, these values are subject to political discretion, as they directly influence tax revenues. Mayors, therefore, have incentives to adjust them strategically based on election cycles. Between 2001 and 2016, assessed values remained stagnant for extended periods, followed by abrupt increases driven by fiscal needs rather than market dynamics.

In contrast, FAR is updated incrementally as new floor space is built and it is less susceptible to political influence. It correlates with land value, as scarcer land (higher prices) is used more intensively (higher FAR). However, FAR grows slowly over time and may fail to reflect real estate booms, as seen in São Paulo during the 2010s. Moreover, it rarely decreases during market downturns, since buildings are long-lived and replaced gradually.

In the first part of the paper, I focus on cross-sectional variation within each year and use FAR as a proxy for land value, as it offers a more accurate snapshot of annual differences across locations. Although direct data on market prices are limited, Appendix Table C.1 provides suggestive evidence that FAR is reasonably correlated with housing prices, at least for cross-sectional comparisons.

In the second part, long-term temporal variation is key, and I instead rely on assessed values for two main reasons. First, while political discretion can freeze assessed values in the short run, periodic adjustments help preserve some of their longer-term trends. Second, I take advantage of significant revisions around 2015, which aimed to partially close the gap between assessed and market values.

[15] For example, baseline values differ between commercial and residential properties and vary based on construction materials and building typology (e.g., multi-family vs. single-family). Additionally, property values are adjusted for depreciation and other factors.

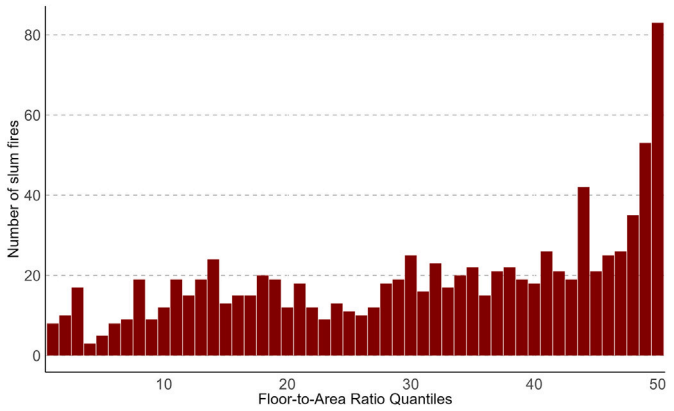


Fig. 2. Number of fires by quantiles of average real formal property value around slums, São Paulo (200–2016).
Notes: Figure depicts the number of fires for each slum-year observation in São Paulo between 2001 and 2016, split by quantiles of Floor-to-Area Ratio. Because quantiles are calculated separately for each year, the same slum can be in different quantiles across years.

Using either FAR or assessed value as proxy, slum land value is computed as shown in Eq. (1).

$$SlumValue^{r}_{st} = \frac{\sum_{i=1}^{I} \mathbb{1}[distance(i,s) \leq r] \times Proxy_{it}}{\sum_{i=1}^{I} \mathbb{1}[distance(i,s) \leq r]} \tag{1}$$

Such that $SlumValue^{r}_{st}$ is the inferred land value of each slum s, $Proxy_{it}$ denotes either the floor-to-area ratio or the assessed value per square meter of formal property i at time t. The set I comprises all formal properties in São Paulo, and $distance(i,s)$ measures the distance in meters between formal property i and the boundary of slum s. Additionally, r defines the buffer radius around each slum s, which is fixed at 500 m. Land value is thus approximated by the average of $Proxy_{it}$, calculated using only formal properties located within r meters of slum s.[16]

3.4. Combining the data

Fig. 2 presents the distribution of slum fires across fifty quantiles of Floor-to-Area Ratio, used as proxy for land value. The number of fires increases with slum land value in a non-linear fashion. While the average number of fires is roughly between 10 and 20 over most of the distribution, it increases sharply in the top quantiles. In particular, the fiftieth quantile is responsible for almost 10 % of all fires in the city.

Although not all fires result from arson, their distribution suggests a discontinuity consistent with strategic arson. Based on Section 2.4, if strategic arson is happening, we should observe a positive probability of arson only after a given land value threshold is exceeded.

The main competing explanation for this pattern is that slums in the top quantiles are disproportionately larger, denser, or have worse infrastructure—all of which could drive a larger number of accidental fires. However, the distributions of variables related to these potential confounders do not follow the same pattern as the fire data. These distributions are smoother, with no clear discontinuities in the top quantiles. Appendix Fig. C.5 illustrates this, suggesting that the unusual behavior of slum fires in São Paulo is not simply a result of poorer infrastructure or higher density.

This pattern of fires also does not appear to be linked to sorting. Using Census data, I compare the evolution of slum density and infrastructure with land value trends between 2000 and 2010. There is little correlation between these variables, suggesting that vulnerability to more accidental fires is not increasing in slums located in increasingly expensive neighborhoods. Appendix Fig. C.6 illustrates this point.

The next sections provide more robust empirical approaches trying to identify the relationship between land value and fires across slums on private and public lands.

4. Cross-sectional approach

4.1. Empirical strategy

The first empirical exercise compares the probability of fire across slums with different land values, but otherwise similar observable characteristics. Following predictions from the theoretical model, I test not only whether fires are positively affected by land value, but also whether this effect is non-linear. I explore variation coming from slums in different quantiles of the land value distribution, approximated by FAR. To add another source of identification, I also compare outcomes across slums on private and public lands.

The main challenge is that no direct measure of arson is available. Instead, the data contain all sorts of fires, including those that are accidental and unrelated to strategic arson. If the incidence of these accidents is also correlated with slum land value, this could lead to omitted-variable bias.[17] For example, slums on more expensive land might be denser because residents want to be closer to job opportunities. This would mechanically increase the probability of fire simply due to more people causing accidents.

I address this concern by including variables that account for heterogeneous fire hazards across slums in different quantiles of land value. Whenever possible, these control variables are split into quantiles to allow for non-linear effects. Hence, identification stems from unconfoundedness conditional on observables. I rely on the assumption that the covariates included in the empirical model account for all factors that cause accidental fires and are correlated with land value.

I also assume there is no sorting of slums with higher probability of fire into higher-value land. The previous section mentioned descriptive evidence corroborating this assumption (in Appendix Fig. C.6). Additionally, slums in São Paulo were largely built prior to 1990, much earlier than the sample period (2001–2016). Hence, slum settlers likely had limited information on the distribution of future land prices across the city at that time.

Even if covariates are insufficient to fully address these concerns, I leverage the fact that legal ownership of slum land creates plausibly exogenous variation in incentives to forcibly remove slums from high-value plots. Specifically, I expect slums on private land to be *a priori* more vulnerable to strategic arson than those on public land. Moreover, land ownership does not seem to be correlated with density or infrastructure, as suggested by Table 1.

Finally, to deal with varying accuracy in the geocoding procedure, as discussed in Section 3.2, not only do I add year fixed effects, but I also calculate the quantiles of land value within each year. This means that almost all of the variation in land value and probability of fire is coming from the cross-section, thus avoiding comparisons of different periods.

Eq. (2) presents the empirical model.

$$\begin{aligned} ProbFire_{st} = & \sum_{k \in [1:K],\, k \neq k_0}^{K} \delta_k \mathbb{1}[QuantileSV_{s,t} = k] \times Private_s \\ & + Private_s + \sum_{k \in [1:K],\, k \neq k_0}^{K} \phi_k \mathbb{1}[QuantileSV_{s,t} = k] \\ & + X_s \times \tau_t + DistrictYear_{st} + \epsilon_{st} \end{aligned} \tag{2}$$

The probability of fire in slum s at time t, $ProbFire_{st}$, is a function of that slum's position in one of the K quantiles of slum land

[16] For further details on this calculation, please refer to Appendix C.4.

[17] Alternatively, one can think of this as a non-classical measurement error problem.

value. This is represented by the sum of dummies $\mathbb{1}[QuantileSV_{s,t} = k]$, indicating whether slum s is at quantile k at time t. The expression $k \in [1 : K]$, $k \neq k_0$ clarifies that one of the quantiles is excluded from the equation to avoid collinearity. $Private_s$ indicates whether a slum is on private land, and interacting it with quantiles yields the differential effect across slums on private and public lands.

Vector X_s is a set of constant slum and neighborhood characteristics observed before 2001 and correlated with accidental fires. I divide the following *pre-period* covariates in the same number of quantiles as the main independent variable: floor-to-area ratio, area of slum in hectares, residents per household, and households per hectare. I add the following covariates without breaking them into quantiles, because there is not enough variation to form unique groups: share of households with access to piped water and waste collection; share of heads of households with monthly income up to 3 times minimum wage; share of heads of households with no income; and share of slum area facing landslide risk.

All covariates are interacted with year fixed effects. I also include District-Year fixed effects to capture localized trends across the 96 districts of São Paulo, as well as to restrict comparisons to slums that are close to each other.

The coefficients of interest are the δ_k and the ϕ_k, which provide insights on three features of the data. First, whether slums in higher quantiles are abnormally affected by fires. Second, whether the non-linearity observed in the descriptive section is robust. Third, whether the effect is stronger for slums on private land.

4.2. Results

Fig. 3 shows estimates for Eq. (2) including all covariates. All errors are robust to spatial correlation, following (Conley, 1999). Panel 3(a) presents the average effect for all slums, regardless of land ownership. The probability of fire seems higher in the highest quantiles, but the effect is rather imprecise and unclear. Panel 3(b) shows that land ownership is key. This regression includes the interaction between land value quantiles and dummies indicating whether the land is private or public.

The probability of fire is significantly different across private and public slums, specifically in the highest quantile of the land value distribution, even after controlling for other factors related to fire hazards. This indicates that strategic arson is happening, but only on private land. There is no evidence of fires related to real estate value on public land. Moreover, Appendix Fig. D.1 shows that results are robust to the exclusion of slums on land whose ownership is unknown.

A similar result is obtained when using quantiles of assessed values instead of FAR. Although noisier, the estimated effect is larger, showing an abnormal probability of fires for slums on private land and in the highest quantile. This is presented in Appendix Fig. D.2.

Additionally, I find no significant differences in covariates across quantiles and land ownership categories, suggesting that accidental fires are not driving the results. In Appendix Fig. D.3, I estimate an analogous model as in Fig. 3(b), replacing the dependent variable with the continuous covariates used to measure slum infrastructure.

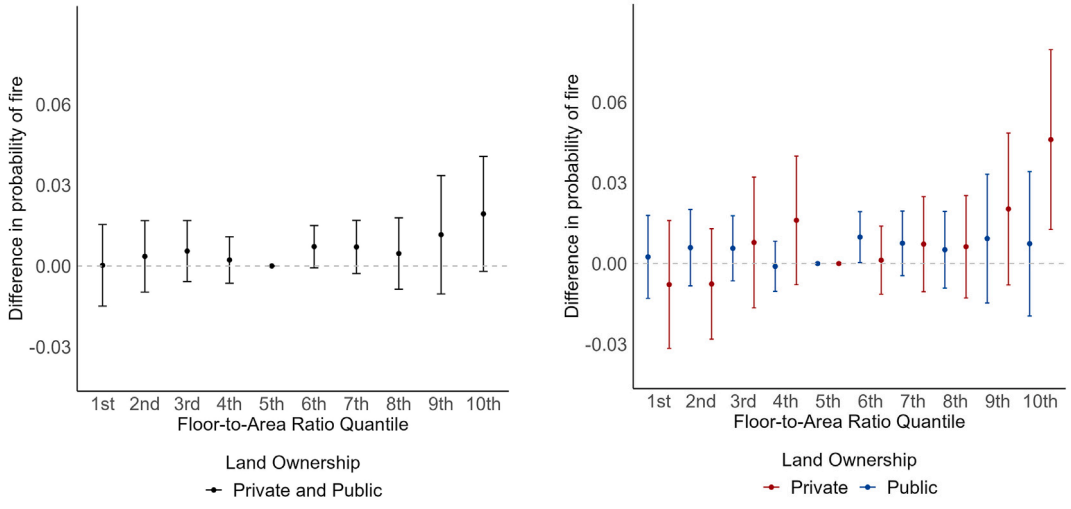
Finally, one potential concern is the choice of how many quantiles to use. I address this by implementing a rolling window estimation to detect discontinuities in the probability distribution of slum fires over land values. Appendix Fig. D.4 shows suggestive evidence of a discontinuous increase in the probability of fire around the 90th percentile, which is only present for slums on private lands. This is consistent with both the main cross-sectional results and the theoretical model predictions.

In summary, there is an abnormal discontinuity in the probability of fire specifically for slums on private land. This does not seem to arise from differences in slum infrastructure, but rather from economic incentives.

5. Difference-in-differences

5.1. Brief context

Cross-sectional estimates suggest that strategic arson may be happening in slums on private land, but not on public land. There might be, however, other concerns about potentially omitted variables driving the results.



(a) All Slums (b) Slums in Private or Public Land

Fig. 3. Estimated difference in probability of fire across floor-to-area ratio quantiles with full set of controls, 2001–2016.
Notes: Panel (a) shows results for a linear probability model with the occurrence of slum fires on the left-hand side and ten quantiles of Floor-to-Area Ratio on the right-hand side. The specification includes district-year fixed effects as well as the following covariates (measured before sample period) interacted with year fixed effects: ten quantiles of area in square meters, ten quantiles of FAR, ten quantiles of households per hectare, ten quantiles of residents per household, share of piped water and waste collection, share of residents with no income, share of residents with low income, and share of slum area facing landslide risk. Panel (b) shows the same regression, but interacts quantiles with the ownership of slum-occupied land. *Private* indicates slums on private areas and *Public* indicates slums on public areas. Errors in both specifications are spatially clustered as in (Conley, 1999), with a 500-meter radius.

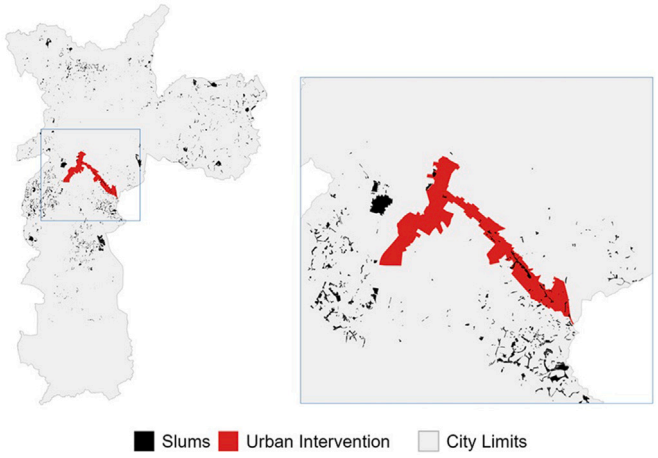


Fig. 4. Urban intervention Água espraiada and slums within four kilometers.
Notes: The map on the left shows the municipality of São Paulo in grey with slums in black. The area in red represents Urban Intervention *Água Espraiada*. The map on the right details slums either inside the intervention zone or within 4 km of its border.

On the one hand, such confounding factors would need to be quite specific. Not only would they be uncorrelated with observable characteristics included in the model, but also simultaneously associated with high land value and land ownership. On the other hand, the cross-sectional analysis does not control for slums' unobservable initial conditions.

To improve identification, I run a Difference-in-Differences strategy leveraging a large shock on land value caused by the *Operação Urbana Água Espraiada* project in São Paulo. This was an urban renovation initiative focused on improving infrastructure and easing zoning restrictions for a set of neighborhoods still a bit far from the city center. The intervention boosted the demand for land, conceivably causing slums in the area to become more valuable for real estate development purposes.

Fig. 4 shows the location of urban intervention *Água Espraiada* in São Paulo, occupying approximately 13 square kilometers, or little less than 1 % of the city's area. The figure also details all slums within a 4-kilometer radius of the intervention. Out of 546 slums in this region, 62 were inside the perimeter of *Água Espraiada*.

The intervention consisted of a series of public auctions starting in 2004. The municipal government sold permits to developers to build above zoning restrictions inside the intervention zone.[18] Revenue would then be invested in urban improvements in the same zone, such as new bridges and parks, more public transit services, and wider avenues. Fig. 5 shows the evolution of permitted extra floor space following the auctions. The amount of floor space quotas exceeding zoning restrictions by the end of 2016 represented 20 % of the intervention zone's land area.

This region was not completely under-served prior to the intervention. Previous projects had already expanded its main avenue during the 1990s, removing some slums, and attracting new people to the area. The new urban intervention came after a relatively stagnant period, bringing visible improvements, especially in road quality and public transit (Nobre, 2009). Hence, land value is expected to have increased not only due to higher demand, but also due to better urban infrastructure and amenities.

Local land market demand reacted positively to the intervention. Both building permit requests and FAR increased following 2004. In 2003, the stock of permit requests within the perimeter of *Água Espraiada* was 50 % lower compared to neighboring areas (within 4 km). In 2007,

[18] The initiative was regulated in 2001 by Municipal Law 13.260, but it was effectively implemented after the first auctions, in 2004.

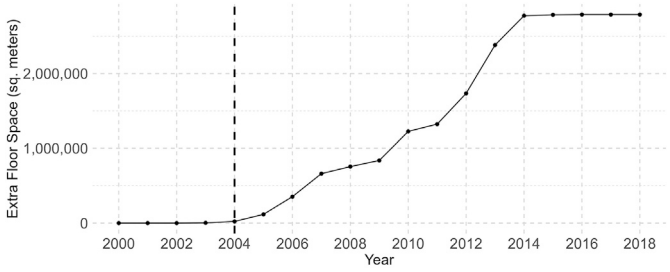


Fig. 5. Stock of additional floor space created inside urban intervention *Água Espraiada*.
Notes: Figure depicts the stock of extra floor space (above standard zoning restrictions) requested by developers and authorized by the municipal government inside *Água Espraiada*. Since it was the 2004 intervention that allowed the creation of extra floor space, the level is zero prior to its implementation.

the stock of requests inside *Água Espraiada* was almost twice as large as the stock outside the intervention. More modestly and slowly, FAR inside *Água Espraiada* went from 10 % below to 5 % above the FAR outside the intervention zone. The two panels in Appendix Fig. D.5 illustrate this change.

5.2. Empirical strategy

With a Difference-in-Differences model, I explore both the timing of *Água Espraiada* intervention and the fact that slums inside its perimeter (treated group) were relatively more exposed to the shock in land value than those farther away from it (control group). I focus on private lands, but refer to results for public lands in the Appendix throughout the text.

In the main sample, the control group includes slums that are farther than 500 m from Água Espraiada and within 4 km of it. These thresholds serve two purposes. First, I discard slums that are too close to the intervention to avoid contamination. Second, I define a maximum distance to keep treated and control groups more comparable. Because São Paulo is a large city, sampling slums from too far away may compromise comparability. Robustness regarding this threshold is discussed in the next section.

The empirical model is given by Eq. (3):

$$ProbFire_{st} = \beta_1 Treated_s \times I_{t\geq 2005} + \lambda_s + \tau_t + X_s \times \tau_t + \varepsilon_{st} \quad (3)$$

As before, $ProbFire_{st}$ is an indicator for whether slum s experienced a fire in year t; λ_s are slum fixed effects, controlling for unobservable initial conditions; X_s is a vector of constant slum and neighborhood characteristics that are interacted with year fixed effects τ_t to control for changes in infrastructure over time. $Treated_s$ identifies slums within the urban intervention zone, while the treatment period is defined by $I_{t\geq 2005}$. Because the intervention began in July 2004 and only annual data are available, I define 2005 as the first full year of treatment.

Coefficient β_1 measures the effect of the policy-driven land value shock on the probability of fire in treated slums compared with the control group. If strategic arson is happening, we should observe a positive and significant estimate for β_1. The main identification assumption is that the probability of strategic arson would have evolved similarly in treated and control slums in the absence of the land value shock.

Although the sample is restricted to slums close to the intervention, characteristics may still differ across units inside and outside of *Água Espraiada*. To improve balance between groups, I employ an Inverse Probability Weighting (IPW) algorithm (Hirano et al., 2003; Busso et al., 2014; Cunningham, 2021). The details of this procedure are described in Appendix D.2.2.

Table 3 shows balance in observable characteristics for both unmatched and matched (IPW) samples for private lands. These characteristics, measured prior to the sample period, are used for estimating propensity scores. In the unmatched sample, there are some baseline

Table 3
Propensity-score matching balance test for slums in private land inside or within 4 km of urban intervention.

	Unmatched			Matched		
	Treated (1)	Control (2)	Difference (3)	Treated (4)	Control (5)	Difference (6)
(log) Neighborhood F.A.R.	−0.335	−0.737	0.402	−0.454	−0.403	−0.051
			(0.063)			(0.065)
Households per hectare	767.317	1715.84	−948.523	1048.497	662.464	386.034
			(582.126)			(193.27)
(log) Slum area	−0.513	−1.034	0.521	−0.957	−0.62	−0.337
			(0.257)			(0.29)
Residents per household	3.813	3.729	0.084	3.75	3.729	0.02
			(0.062)			(0.063)
Share income up to 3 salarios minimos	0.454	0.405	0.05	0.411	0.405	0.006
			(0.023)			(0.027)
Share of no income	0.157	0.162	−0.005	0.146	0.157	−0.011
			(0.02)			(0.021)
Share landslide risk	0.027	0.203	−0.175	0.008	0.009	−0.001
			(0.061)			(0.014)
Share waste collection	0.983	0.979	0.005	0.991	0.986	0.005
			(0.01)			(0.008)
Share water	0.991	0.982	0.009	0.994	0.992	0.002
			(0.008)			(0.005)
Observations	28	137	-	25	31	-

Notes: Columns (1) and (2) show the means of each variable conditional on treatment status. Columns (4) and (5) show the re-weighed means such that treated and control slums receive weights 1/p and 1/(1-p) respectively. Columns (3) shows the estimated coefficient of an OLS regression of each variable on treatment. Columns (6) repeats the regression in Column (3), but re-weighing according to (4) and (5).

Table 4
Effect of urban intervention on strategic arson in slums on private land.

Dependent variable:	Probability of fire			
Model:	(1)	(2)	(3)	(4)
Inside OUC	−0.07* (0.04)			
Year ≥ 2005	−0.03 (0.02)	−0.03 (0.02)		
Inside OUC × Year ≥ 2005	0.06** (0.03)	0.06** (0.03)	0.06** (0.03)	0.07*** (0.02)
Slum FE (56)		Yes	Yes	Yes
Year FE (15)			Yes	Yes
IPW covariates * Year				Yes
Observations	840	840	840	840
R^2	0.008	0.357	0.370	0.523
Within R^2		0.007	0.007	0.248

Notes: Slums located inside or within 4 km of Urban Intervention. Inverse Propensity Score Weighing (IPW) applied to all specifications such that treated units receive 1/p weights whereas control ones receive 1/(1-p) weights. Column (1) has no controls. Column (2) adds slum fixed effects; Column (3) adds year fixed effects; Column (4) adds interactions between IPW covariates and year fixed effects. Spatially robust errors are based on Conley (1999) with 500-meter distance threshold. *p<.1; **p<.05; ***p<.01.

differences, such as in Neighborhood FAR.[19] After re-weighing observations and trimming propensity scores with extreme values, I obtain a more comparable sample. Appendix Fig. D.6 shows the distribution of propensity scores for treated and control slums on private and public lands.

5.3. Results

I first analyze the effects of the urban intervention on the probability of fire. Then, I show that these results are likely driven by slum land value.

5.3.1. Fires

Table 4 shows results using the Difference-in-Differences model with IPW for private slums. This sample considers only observations within 4 km of the urban intervention's border.

The effect of the land value shock driven by *Água Espraiada* on the probability of slum fire is both positive and significant. Moreover, estimates are fairly stable after the inclusion of slum and year fixed effects. They are also robust to adding the interaction of year fixed effects with covariates.[20] All errors are robust to spatial correlation (Conley, 1999).

Appendix Table D.4 presents the same specifications for the unmatched sample (without IPW). In this case, results are sensitive to the inclusion of covariates. Once they are included, the estimate is comparable to the one from the matched sample. This is likely driven by baseline imbalances in the unmatched sample, particularly for some covariates such as the logarithm of FAR in the pre-period or share of area facing landslide risk. With IPW, imbalances are smoothed and results become more stable. There is no evidence of pre-trends, as shown in Fig. 6. Because the sample is small, there is insufficient variation to estimate all annual coefficients. Moreover, standard errors can be quite large because fire is a rare event. Overall, however, point-wise estimates are positive after 2004.

There might be concerns about limiting the distance threshold to four kilometers around the urban intervention. Fig. 7 shows that changing this parameter does not affect estimates substantially. The first two

[19] Appendix Table D.3 shows the same statistics for slums on public land.

[20] These are the same covariates as the ones included in the propensity score estimation. Covariate levels are fixed to pre-treatment period.

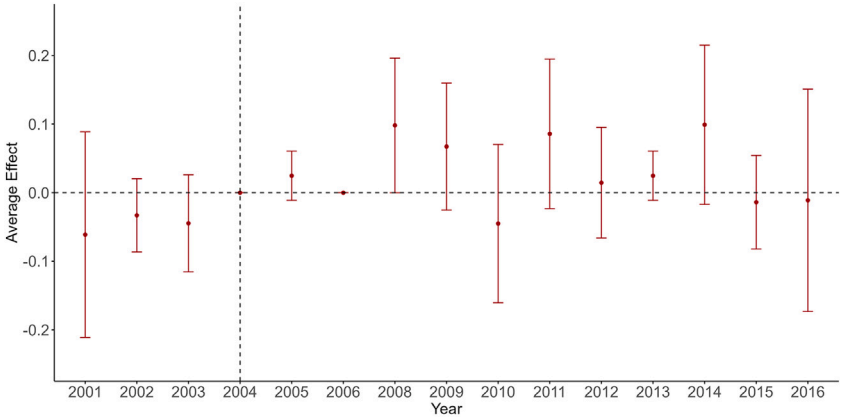


Fig. 6. Annual effect on probability of fire for slums on private land.
Notes: Because the sample is small, there is not enough variation to estimate all annual coefficients. There is no data on fires in 2007, so this year was excluded from the sample. Errors are spatially clustered as in Conley (1999) with a 500-meter radius.

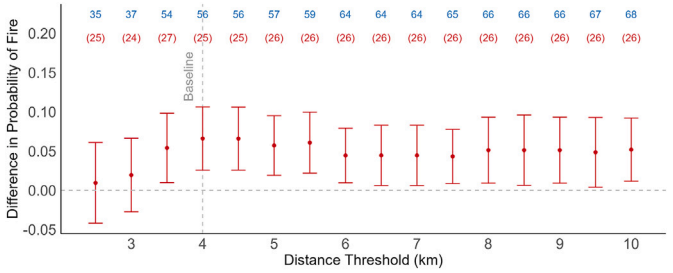


Fig. 7. Effect on probability of fire for each maximum distance threshold - slums on private land.
Notes: Figure depicts estimates of the effect of the urban intervention on the probability of fire for slums on private land. Each estimate corresponds to a sample of slums that are within a certain distance of the urban intervention. This maximum distance is indicated by the x-axis. Because of the IPW procedure, the number of observations in treated and control groups varies. The total number of observations for each maximum distance is shown in blue, whereas the number of treated units is shown in red and in brackets. Treated units are all slums inside the perimeter of the urban intervention. Control units are all slums located farther than 500 m from the urban intervention and within each distance threshold. Units with estimated propensity score higher than 0.95 or lower than 0.05 were discarded.

categories are an exception (2.5 and 3 km) because there is a relatively small number of control units. There are only 10 and 13 slums in the control group within the 2.5 km and 3 km thresholds, respectively, compared with more than 20 for all other thresholds. Therefore, with a reasonably sized control group, estimates seem to be robust.

Appendix Fig. D.7 replicates this sensitivity analysis for the unmatched sample of private slums. For all thresholds, estimated effects are very close to the ones from the matched sample. Estimates are significant at smaller maximum distances but lose significance as the distance increases—likely because more distant control slums are less comparable to treated ones.

Different geocoding success rates for fires over time may also pose an issue. To alleviate this concern, apart from including year fixed effects, I show in Appendix D that results hold for multiple sub-samples. Table D.5 shows similar effects for shorter sample periods.[21] Furthermore, Fig. D.8 illustrates that results are robust to combinatorial pair-wise exclusions of years from the sample.

Another concern is that covariates might not fully account for confounding factors evolving over time, such as slum density. I address this possibility by examining what happens with slums on public land. If density is increasing because of the intervention, this should affect all slums, not just the ones on private land. Table 5 shows no effect of the intervention for slums on public land. Excluding slums on land with unknown ownership leads to a similar conclusion, as shown in Appendix Table D.6.

In summary, estimates are positive and significant only for slums on private land. This highlights the importance of land ownership and provides more evidence on strategic arson. If fires were simply related to infrastructure or density, we would observe similar results for private and public lands, both in the cross-sectional and Difference-in-Differences approaches.

5.3.2. Land value

In this section, I provide some evidence that the abnormal increase in the probability of fires is linked to higher land values within the intervention zone. In this analysis, I use assessed values instead of FAR. Given the importance of temporal variation, FAR may be less suitable because it is persistent, evolves slowly, and may exhibit less variation in more established areas.

To study the impact on land value, I rely on the fact that it reflects expectations about future housing prices—developers base their decisions on anticipated profits from selling new units built on vacant plots. I therefore use long-term changes in assessed values to infer shifts in land value over time.

Focusing on long-term price trajectories also offers two key advantages. First, it mitigates a primary concern with assessed values: their infrequent updates due to political discretion. Second, it allows me to incorporate a major reassessment during the 2013–2016 electoral cycle, when the municipal government adjusted assessed values to bring them closer to market prices.[22]

To capture long-term trajectories, I run a two-period Difference-in-Differences with IPW, restricting the sample to 2004 (the last pre-treatment year) and 2016 (the most recent available year). Table 6 presents the effect of the intervention on assessed value per square meter around slums. It shows results for all slums (Columns 1–4) and for those

[21] The shortest sample is restricted to the 2001–2006 period, which falls within a single data group according to Fig. C.3.

[22] For instance, assessed values increased by an average of 40 % in a single year.

Table 5
Effect of urban intervention on strategic arson in slums on public land.

Dependent variable:	Probability of fire			
Model:	(1)	(2)	(3)	(4)
Inside OUC	0.02 (0.03)			
Year ≥ 2005	0.007 (0.009)	0.007 (0.009)		
Inside OUC × Year ≥ 2005	−0.01 (0.02)	−0.01 (0.02)	−0.01 (0.02)	−0.02 (0.03)
Slum FE (101)		Yes	Yes	Yes
Year FE (15)			Yes	Yes
IPW covariates * Year				Yes
Observations	1515	1515	1515	1515
R^2	0.002	0.193	0.207	0.330
Within R^2		0.000	0.000	0.155

Notes: Slums inside or within 4 km of Urban Intervention. Inverse Propensity Score Weighing (IPW) applied to all specifications such that treated units receive 1/p weights whereas control ones receive 1/(1-p) weights.Columns (1) has no controls. Column (2) adds slum fixed effects; Column (3) adds year fixed effects; Column (4) adds interactions between IPW covariates and year fixed effects. Spatially robust errors based on Conley (1999) with 500-meter distance threshold. *p<.1; **p<.05; ***p<.01.

Table 6
Effect of urban intervention on assessed values per square meter around slums.

Dependent variable:	(log) Assessed value							
	All slums				Private land			
	Unmatched		Matched		Unmatched		Matched	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Inside OUC	0.15**		−0.03		0.13**		−0.07	
	(0.07)		(0.05)		(0.06)		(0.06)	
Year ≥ 2005	0.64***		0.74***		0.62***		0.78***	
	(0.03)		(0.03)		(0.04)		(0.06)	
Inside OUC × year ≥ 2005	0.30***	0.17***	0.13**	0.13***	0.24***	0.11**	0.04	0.07**
	(0.06)	(0.04)	(0.05)	(0.03)	(0.04)	(0.04)	(0.07)	(0.03)
Slum FE		Yes		Yes		Yes		Yes
Year FE (2)		Yes		Yes		Yes		Yes
IPW covariates * year		Yes		Yes		Yes		Yes
# Slum FE	–	494	–	203	–	165	–	56
Observations	988	988	406	406	330	330	112	112
R^2	0.625	0.969	0.704	0.984	0.653	0.978	0.768	0.989
Within R^2		0.496		0.603		0.601		0.680

Notes: Slums located inside or within 4 km of Urban Intervention. Inverse Propensity Score Weighing (IPW) applied to all specifications such that treated units receive 1/p weights whereas control ones receive 1/(1-p) weights. Columns (1)–(4) show estimates for all slums, on either public or private lands. Columns (5)–(8) show estimates only for slums on private land. Odd-numbered columns have no fixed effects or other covariates. Even-numbered columns include slum and year fixed effects, as well as covariates. Columns (1), (2), (5), and (6) show estimates based on unmatched samples. Columns (3), (4), (7), and (8) show estimates using IPW to match observations. Spatially robust errors are based on Conley (1999) with 500-meter distance threshold. *p<.1; **p<.05; ***p<.01

on private land (Columns 5–8), using both unmatched and matched samples.

Results indicate that slums within *Água Espraiada* experienced a 13 % rise in surrounding property values from 2004 to 2016, signaling that land in the area became more expensive. I find positive estimates for slums on private land, but coefficients are more sensitive to the inclusion of covariates.

Combined with previous findings, these results suggest that land value increased across the board, but only slums on private land faced a higher probability of fire. This pattern aligns with the idea that incentives for strategic arson are stronger among private landowners.

Finally, these estimates help reconcile the cross-sectional and Difference-in-Differences approaches. The increase in land value driven by the intervention seems large enough to push slums across the threshold implied by the cross-sectional analysis. In particular, shifting the average slum from the 9th to the 10th quantile would require a 6 %–7 % increase in land value, which is close to the estimated effects in Table 6.

6. Conclusion

This paper investigates whether land conflicts also arise in urban settings, albeit in a different form. Instead of open confrontations between armed groups, these conflicts may manifest through arson targeting slums as a violent means of displacement.

To overcome challenges such as measurement error and omitted variable bias, I run two empirical exercises: a cross-sectional analysis and a Difference-in-Differences. In both, I show evidence that fires are abnormally more likely to happen in slums with high land value. This effect is exclusive to slums on private land and it does not seem to be driven by accidents. Moreover, both approaches are consistent with each other and with theoretical predictions. The urban intervention studied in the Difference-in-Differences seems to move slum land value across the threshold implied by cross-sectional estimates.

Such findings corroborate the hypothesis that arson might be a violent tool in urban land conflict. These results also highlight the importance of addressing the lack of land property rights in slums. As is the case in rural areas, disputes between residents and landowners can escalate to violence if enforcement of land property rights is imperfect.

To compensate for this, governments could take a more active role in increasing the cost of violence, for instance, by pushing to improve slums' infrastructure and land ownership status. Moreover, part of this issue might also be mitigated by policies that reduce the creation of new slums.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work I used ChatGPT to check grammar and cohesion of some paragraphs. After using this tool/service, I reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Anecdotal evidence from news pieces

The following list contains some cases of fires associated with slum removal episodes in high-value neighborhoods, in Brazil and other countries.

1. Fire during removal in Praia do Pinto (Rio de Janeiro, RJ) – 1969
   http://www.encontro2012.historiaoral.org.br/resources/anais/3/1339790201_ARQUIVO_MemoriasdaRemocaoABHO2012.pdf
2. Fire in Ocupacao Nelson Mandela (Osasco, SP) – 2015
   https://sao-paulo.estadao.com.br/noticias/geral,moradores-ateiam-fogo-em-barracos-durante-reintegracao-de-posse-em-sp,1702643
3. Fire in Favela Estaiadinha (São Paulo, SP) – 2011
   http://g1.globo.com/sao-paulo/noticia/2011/09/moradores-sao-retirados-de-favela-durante-reintegracao-de-posse-em-sp.html
4. Fire in Favela Real Parque (São Paulo, SP) – 1992, 2010
   https://rollingstone.uol.com.br/edicao/56/arquitetura-da-destruicao/
5. Fire in Favela Buraco Quente (São Paulo, SP) – 2004, 2012, 2014
   https://www.redebrasilatual.com.br/cidadania/2014/09/incendio-na-favela-do-buraco-quente-terminou-o-servico-de-alckmin-contra-moradores-7676.html
6. Fire in Ocupacao Esperanca (Osasco, SP) – 2016
   http://agenciabrasil.ebc.com.br/geral/noticia/2016-09/incendio-em-area-desapropriada-causa-estranheza-em-prefeito-de-osasco
7. Fire in New Delhi, India – 2018
   https://www.Theguardian.com/cities/2018/sep/05/devastated-destroyed-delhi-slums-recover-fires
8. Fire in Dhaka, Bangladesh – 2016
   https://archive.dhakatribune.com/bangladesh/2016/12/24/dhaka-slum-fires-arson
9. Fire in Favela do Piolho (São Paulo, SP) – 2014
   https://sao-paulo.estadao.com.br/noticias/geral,bombeiros-acreditam-que-incendio-em-favela-de-sp-foi-criminoso,1556959
10. Fire in Favela do Cimento (São Paulo, SP) – 2019
    https://www1.folha.uol.com.br/cotidiano/2019/03/incendio-toma-conta-de-favela-na-radial-leste-que-seria-desocupada.shtml
    https://g1.globo.com/sp/sao-paulo/noticia/2019/03/25/policia-prende-suspeito-de-iniciar-fogo-que-destruiu-a-favela-do-cimento-no-entorno-do-viaduto-bresser.ghtml
11. Fire in Favela 21 de Abril (São Paulo, SP) – 2014
    http://g1.globo.com/bom-dia-brasil/noticia/2014/10/um-em-cada-quatro-incendios-em-favelas-de-sao-paulo-e-criminoso.html

Appendix B. Theoretical model

B.1. Basic structure

The model focuses on the decision of a landowner who claims ownership of a squatted land plot. I assume that a repossession claim is already running in court and she is now waiting for a decision, which could take years. Uncertainty about the timing of a court ruling and its outcome, combined with increasing land value, poses a dynamic trade-off to the landowner: she can either wait another period for a court decision; or she can cause the slum to burn, anticipating the conclusion of the dispute. By doing this, she either gains control over the land or goes to prison with some probability.

This theoretical approach was originally used to explain landowners' decisions to develop empty land plots based on housing prices (Capozza and Li, 1994; Capozza and Sick, 1994). The landowner would compare the potential gains from development with those of an outside option, such as a constant agricultural income. As housing becomes more expensive, developing an empty land plot becomes increasingly more attractive. Eventually, prices hit a threshold beyond which it is unequivocally more profitable to develop the plot. When this happens, the landowner irreversibly decides to convert the land plot into a building.

The decision to burn a slum entails some additional features. First, not only does it depend on the value of a reclaimed land plot, but also on how likely the court is to provide a decision on the matter. Second, it also hinges on how likely the decision is to favor the landowner. Third, it depends on the moral and financial costs of committing arson, as well as on the probability of destroying the slum in the process. Finally, waiting for courts imposes legal costs, such as lawyers' hourly rates for example.

Analogously to the canonical case, in every period of time t the agent faces a trade-off between *waiting* for another period or taking irreversible action to interrupt the slum occupation. Behind this decision is the value of the slum-occupied land i, p_{it}, which determines the landowner's utility, $u(p_{it})$, in case of slum removal.

Price p_{it} can be more or less relevant depending on whether slum-occupied land is private or public. If it is private, p_{it} impacts directly what the landowner gains from the eviction. In the case of public lands, p_{it} might determine the size of the externality generated by slum removal to neighbors. Much like an urban amenity, slum removal would likely be more valuable in neighborhoods where housing or land prices are higher.[23] It is likely, however, that this mechanism produces weaker incentives compared with the private case, simply because there is a coordination problem among neighbors.

With some slight changes, it is possible to allow for weaker incentives in the case of slum removals on public lands. One could, for instance, define function $u_{public}(p_{it}) = (1+\xi)u(p_{it})$ as the utility of removal of slums on public land, such that $\xi \in [-1, 0)$. Discounting factor ξ would then dictate the heterogeneous likelihood of arson in public versus private lands. To simplify the model, I abstract from this additional structure and assume that the baseline case is that of a slum on private land, such that its removal provides utility $u(p_{it})$ for the landowner.

The following is drawn from a basic approach to optimal stopping models, found in Dixit and Pindyck (1994). Formally, the landowner faces a maximization problem represented by the following value function $w(\cdot)$:

$$w(R_{it}) = \max\{u(R_{it}); -c(1-\delta) + \delta E_R[\lambda\theta\Delta v(R_{i,t+1}) + (1-\lambda\Delta)w(R_{i,t+1})]\} \tag{B.1}$$

Such that $R_{it} = log(p_{it})$; Δ is the length of the discrete time interval (i.e. one year); and $\delta = e^{r\Delta}$ represents the discount factor for a given length of the time interval Δ. Moreover, E_R is an expected value and it depends on the realization of the stochastic quantity R_{it}.

If the agent decides to burn the slum, this is irreversible and she gets $u(R_{it})$. This is the first element in Eq. (B.1)'s $\max(\cdot)$ function. Alternatively, the second element represents the decision to wait. In this case, the agent pays legal costs c to maintain her claim in court. With probability λ, courts rule a decision in $t+1$, which favors the landowner

[23] To clarify this point, one could think of the opposite side: in a poor neighborhood, slum and non-slum communities are more similar, and thus removing the former would not significantly change the neighborhood's environment from the perspective of the latter.

with probability θ and yields a payoff of $v(R_{i,t+1})$. With probability $(1-\lambda)$, courts delay their ruling, and the agent confronts the same maximization problem in the future. This postponement is represented by the value function in $t+1$, $w(R_{i,t+1})$.

In summary, the landowner chooses every period whichever is higher between the present payoff of burning the slum and the expected payoff of waiting for a court ruling. This ultimately hinges on how high the real estate value p_{it} is for the occupied land. As one may anticipate, this problem yields an optimal threshold p^*, above which strategic arson is profitable.

I assume a standard dynamic for the real estate prices (Capozza and Li, 1994; Capozza and Sick, 1994). Returns to real estate investment dR follows a Brownian Motion with mean growth rate μ, standard deviation σ and with shocks dz from a standard Wiener process, i.e.[24]

$$dR = \mu\Delta + \sigma dz \tag{B.2}$$

$$z \sim N(0,1) \tag{B.3}$$

B.2. Solution for arson decision

To find a closed-form solution, it is more convenient to work with continuous time. To go from a discrete to a continuous model, I make $\Delta \to 0$ and combine B.1 with B.2. This yields the following second-order differential equation for the continuation region (when the agent decides to wait), analogously to Capozza and Li (1994) and Capozza and Sick (1994):

$$(\lambda + r)w(R) = -rc + \lambda\theta v(R) + \mu w'(R) + \frac{1}{2}\sigma^2 w''(R) \tag{B.4}$$

Such that r is a continuous discount factor.

The solution to this problem is characterized not only by the unique value function $w(\cdot)$, but also by threshold R^*. This in turn defines a p^* that separates the *burn* and *wait* decisions. I find these two quantities by resorting to classical conditions presented in Dixit and Pindyck (1994):

1. **No Dynamics:** when $R \to -\infty$ (or $p \to 0$). In this case, the path of prices over time does not matter anymore. In all time periods, the agent sticks to either *burning* or *waiting*—whichever provides higher utility:

$$w(-\infty) = \max\left\{\frac{-rc + \lambda\theta v(-\infty)}{\lambda + r}; u(-\infty)\right\} \tag{B.5}$$

2. **Value Matching:** the utility of *waiting* and *burning* must be the same at the threshold point R^*.

$$w(R^*) = u(R^*) \tag{B.6}$$

3. **Smooth Pasting:** intuitively, this condition can be thought of as a maximization problem in which, given the value function, the agent chooses the threshold R^* that gives her the highest utility.

$$w'(R^*) = u'(R^*) \tag{B.7}$$

Furthermore, for a solution to exist, the distance between the payoffs of *burning* and *waiting* must decrease with prices. Otherwise, the agent would never decide to burn if she started in the waiting region. Formally, the following expression must be increasing in R:

$$u(R) - (\lambda + r)w(R) + rc - \lambda\theta v(R) - \mu w'(R) - \frac{1}{2}\sigma^2 w''(R) \tag{B.8}$$

[24] Notice that $dR = \frac{dP}{P}$, the return on a real estate asset with price P.

B.3. Defining payoffs

A closed-form solution also requires that I define functional forms for $v(R)$ and $u(R)$. I start with $v(R)$. Resorting to the original problem in the literature, I define $v(R)$ as the payoff for an owner deciding when to develop an empty land plot. I assume no operating costs to maintain the plot undeveloped and the same discrete discount rate, δ.

The landowner receives the discounted future payoffs represented by $v(R')$ if she decides not to develop the land plot, which is the value function in the next period. Alternatively, if she decides to develop the plot, she gets a payoff equal to real estate value e^R minus investment cost I. Formally,

$$v(R) = \max\{e^R - I; \delta v(R')\} \tag{B.9}$$

Under similar conditions as before, the solution is characterized by a unique value function $v(\cdot)$ and threshold R^*, above which the landowner chooses to develop the plot. Assuming returns maintain the same stochastic process as before, I solve the following homogeneous second order differential equation:

$$\frac{1}{2}\sigma^2 v''(w) + gv'(R) - rv(R) = 0 \tag{B.10}$$

subject to analogous conditions:

1. **No Dynamics:** $v(-\infty) = 0$
2. **Value Matching:** $v(\hat{R}) = e^{\hat{R}} - I$
3. **Smooth Pasting:** $v'(\hat{R}) = e^{\hat{R}}$

Assuming that $e^R - I - \frac{1}{2}\sigma^2 v''(R) - gv'(R) + rv(R)$ is increasing in R, I find

$$v(R) = \frac{I}{\alpha - 1}e^{(R-\hat{R})\alpha} \tag{B.11}$$

Such that the positive root of the polynomial is given by

$$\alpha = \frac{1}{\sigma}\left[-\frac{\mu}{\sigma} + \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right] \tag{B.12}$$

and the optimal threshold in the land-development problem is given by

$$\hat{R} = log\left[\frac{\alpha I}{\alpha - 1}\right] \tag{B.13}$$

As for $u(R)$, I define it as the utility of *burning*, based on $v(R)$. However, there is an additional fixed cost k for burning the slum and a probability β of being convicted for arson. In the latter case, the landowner accrues no value from the removal. I.e.,

$$u(R) = (1 - \beta)v(R) - k \tag{B.14}$$

B.4. Closed form solution

We can solve Eq. (B.4) by plugging in Eqs. (B.11) and (B.14), using conditions B.5 through B.8. After some algebra, we get

$$w(R) = e^{(R-R^*)\gamma}\left[\frac{rc}{\lambda + r} + (1 - \theta - \beta)\frac{Ie^{(R^*-\hat{R})\alpha}}{\alpha - 1} - k\right] - \frac{rc}{\lambda + r} + \frac{\theta I}{\alpha - 1}e^{(R-\hat{R})}\alpha \tag{B.15}$$

In which γ is the positive root of the differential equation in B.4

$$\gamma = \frac{1}{\sigma}\left[-\frac{\mu}{\sigma} + \sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda + r)}\right] \tag{B.16}$$

And

$$R^* = \hat{R} + \frac{1}{\alpha} log\left[\frac{\gamma(\alpha-1)}{I(1-\theta-\beta)(\gamma-\alpha)}\left(k - \frac{rc}{\lambda+r}\right)\right] \tag{B.17}$$

Finally, plugging $R = log(p)$ yields an expression in terms of real estate value instead of returns:

$$p^* = \hat{p}\left[\frac{\gamma(\alpha-1)}{I(1-\theta-\beta)(\gamma-\alpha)}\left(k - \frac{rc}{\lambda+r}\right)\right]^{\frac{1}{\alpha}} \tag{B.18}$$

Before moving to the implications of Eq. (B.18), there are two remarks about parameters. First, using B.8 one can show that $k > \frac{rc}{\lambda+r}$. This means that, in terms of costs, the agent prefers using the legal system instead of burning a slum. Second, for the solutions presented here to be well-defined, one needs $\theta + \beta < 1$. This means that the probability of being convicted for arson plus the probability of winning a lawsuit cannot be too high.

Since the agent burns the slum whenever $p \geq p^*$, the probability of observing arson as a function of the slum's real estate value is given by

$$\begin{aligned} P(arson) &= P(p \geq p^*) \\ &= P\left\{p \geq \hat{p}\left[\frac{\gamma(\alpha-1)}{I(1-\theta-\beta)(\gamma-\alpha)}\left(k - \frac{rc}{\lambda+r}\right)\right]^{\frac{1}{\alpha}}\right\} \end{aligned} \tag{B.19}$$

Eq. (B.19) provides some insights into how the probability of arson responds to institutional changes. For example, increasing the probability of either winning a repossession lawsuit (θ) or capturing the arsonist (β) increases the threshold p^*, thus decreasing the overall probability of arson, *ceteris paribus*. Analogously, increasing the cost of arson (k) or decreasing the cost of legal lawsuit (c) would both contribute to less arson cases.

The optimal threshold for developing the land plot ($\hat{p}$) is also important. Increasing $\hat{p}$ means that the landowner would postpone the decision to develop a new building in that plot. Therefore she would be more willing to wait for a court decision rather than burning the slum. Conversely, if $\hat{p}$ is low—and suppose now $\hat{p} < p^*$ —, then the landowner would have developed the plot already in the absence of the slum, which makes her more willing to commit arson.

B.5. Probability of arson and real estate prices

The last step is to find out how the probability of arson responds to an increase in the slum's real estate price. Let us assume that the cost of burning a slum is not the same for all potential arsonists, but instead follows a generic distribution such as

$$k = \tilde{k} \sim F(\cdot) \tag{B.20}$$

Then, isolating $\tilde{k}$ and re-writing Eq. (B.19) gives

$$\begin{aligned} P(arson) &= P\left\{\tilde{k} \leq \left[\frac{p}{\hat{p}}\right]^{\alpha}\left[\frac{I(1-\theta-\beta)(\gamma-\alpha)}{\gamma(\alpha-1)}\right] + \frac{rc}{\lambda+r}\right\} \\ &= F\left(\left[\frac{p}{\hat{p}}\right]^{\alpha}\left[\frac{I(1-\theta-\beta)(\gamma-\alpha)}{\gamma(\alpha-1)}\right] + \frac{rc}{\lambda+r}\right) \end{aligned} \tag{B.21}$$

Differentiating this probability with respect to p and defining $F'(\cdot) = f(\cdot) > 0$ as the probability density function then yields

$$\begin{aligned} &\frac{\partial P(arson)}{\partial p} = \\ &= f\left(\left[\frac{p}{\hat{p}}\right]^{\alpha}\left[\frac{I(1-\theta-\beta)(\gamma-\alpha)}{\gamma(\alpha-1)}\right] + \frac{rc}{\lambda+r}\right) \\ &\alpha\left[\frac{p}{\hat{p}}\right]^{\alpha-1}\left[\frac{I(1-\theta-\beta)(\gamma-\alpha)}{\gamma(\alpha-1)}\right] \end{aligned} \tag{B.22}$$

Since $\theta + \beta < 1$ and $\alpha > 1$ for $\hat{R}$ to be well-defined in Eq. (B.13), demonstrating that $\gamma - \alpha > 0$ implies that this derivative is positive. I do this by plugging the actual values of polynomial roots from Eqs. (B.12) and (B.16) and simplifying the resulting expression:

$$\begin{aligned} \gamma - \alpha &= \frac{1}{\sigma}\left[-\frac{\mu}{\sigma} + \sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda+r)}\right] - \frac{1}{\sigma}\left[-\frac{\mu}{\sigma} + \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right] \\ &= \frac{1}{\sigma}\left[\sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda+r)} - \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right] \\ &= \frac{1}{\sigma}\left[\sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda+r)} - \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right] * \\ &\quad * \frac{\left[\sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda+r)} + \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right]}{\left[\sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda+r)} + \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right]} \\ &= \frac{2\lambda}{\left[\sqrt{\frac{\mu^2}{\sigma^2} + 2(\lambda+r)} + \sqrt{\frac{\mu^2}{\sigma^2} + 2r}\right]} > 0 \end{aligned} \tag{B.23}$$

Hence, increasing the real estate value of a slum leads to a higher probability of strategic arson against it.

Appendix C. Data details

C.1. Formal city growth around slums

Fig. C.1 presents the distribution of slums according to the proportion of high-income households living around them. Although most slums are in relatively poor neighborhoods, roughly 20 % of them are in rich areas.

Fig. C.2 shows FAR growth from 2001 to 2016 around slums according to their distance to São Paulo city center (a proxy for the theoretical Central Business Center). Overall, FAR expands across the city, but there is faster growth in some areas far from downtown. This reflects the expansion of the city into more peripheral areas, which could lead to higher land values and pressure to remove slums.

C.2. Geocoding success rates

Fig. C.3 presents the number of slum fires in each year, as well as the three breaks in the data. Above each bar, it shows the geocoding success rate for each year. This is given by the number of slum fires successfully geocoded divided by the total number of slum fires in the state of São Paulo,[25] i.e., the total number of slum fires before geocoding the observations.

One concern is that missing municipal names could make geocoding success rates in group (ii) lower than in the others. This, however, does not seem to be the case when comparing groups (i) and (ii). Indeed, the success rate in geocoding is fairly comparable, at around 10 %, from 2001 to 2010.

Conversely, accuracy increases from 2011 through 2016. During this period, exact residential addresses are omitted in the data. However, the fire department systematically posted the locations of slum fires on their social media accounts.

C.3. Cross-sectional correlation between FAR and market prices

This analysis relies on third-party data on the prices of new buildings in São Paulo from 2001 to 2013.[26]. Due to the sparse distribution of new developments, the data are aggregated at district level. São Paulo comprises 96 districts, each considerably larger than individual slums

[25] I have to consider slum fires in the entire state because the names of municipalities are missing for group (ii).

[26] This is provided by EMBRAESP and it is available at https://centrodametropole.fflch.usp.br/en/download-de-dados.

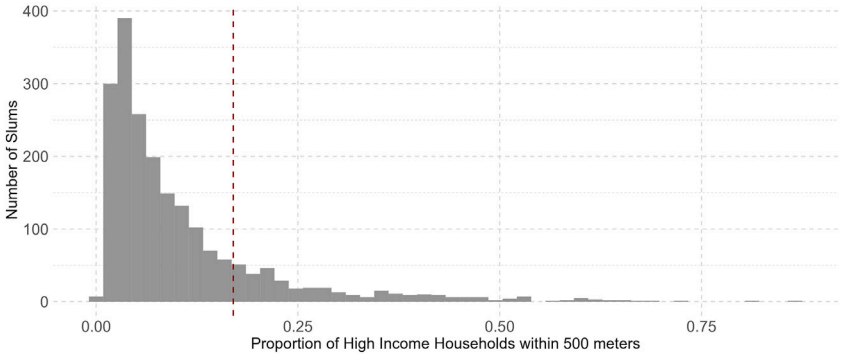


Fig. C.1. Distribution of slums according to proportion of high income household heads around them in São Paulo (2000).
Notes: Figure shows the number of slums according to the percentage of high income households living within 500 m of those slums. Data are from the 2000 Census. High income household heads are defined here as those earning more than 10 times the Brazilian minimum wage. The overall city average proportion of high income households is represented by the vertical dashed red line.

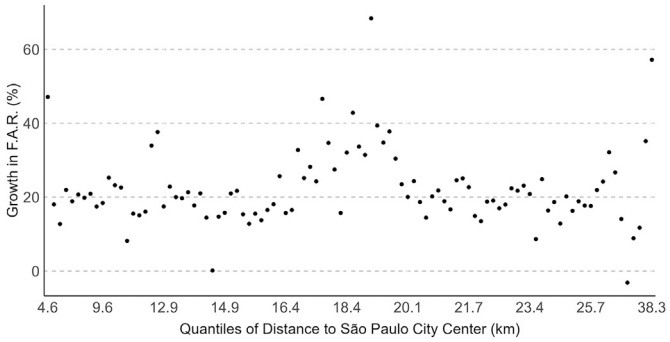


Fig. C.2. Growth in floor-to-area ratio (f.a.r.) between 2001 and 2016 and distance to São Paulo city center.
Notes: Growth in F.A.R. is the percentage change in floor-to-area ratio between 2001 and 2016 for all slums in the sample. Distance to City Center refers to each slum's distance to the historical city center of São Paulo. Just for reference, Quantiles 1 and 100 correspond to 4.6 and 38.3 km on the x-axis, respectively.

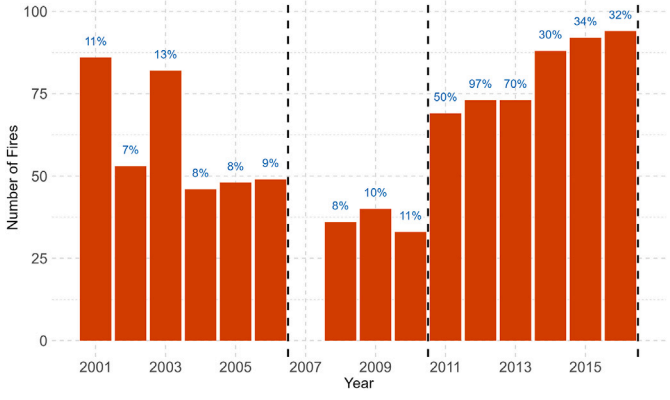


Fig. C.3. Slum fires in São Paulo from 2001 to 2016 and changes in data structure.
Notes: Figure depicts the number of geocoded slum fires in each year. The percentages in blue on top of each bar represent a measure of accuracy for the procedure of geocoding slum fires. They indicate the proportion of geocoded fires in sample compared with total number of fires in the state of São Paulo in each year.

and their immediate surroundings. Unfortunately, the dataset does not extend beyond 2013.

Table C.1 presents the estimated correlation between the floor area ratio (FAR) and the prices of new residential and commercial units during this period. While the correlation fluctuates over time, it remains consistently strong—averaging around 0.6. These results suggest that FAR can serve as a reasonable proxy for land value in cross-sectional analyses.

C.4. Aggregating property tax data around slums

In the main text, the formula for calculating the land value proxy assumes data is available for each property i individually. One limitation, however, is that I observe properties at the street section level, each of which has roughly 200 m. This adds an additional step to calculations, since I must compute an average quantity for each street section and then calculate $SlumValue^{r}_{st}$ by averaging streets at a given distance r from slum s.

When calculating average values within a given radius of slum borders, I account for the fact that some street segments are partially outside these buffer areas. Ignoring this feature could be a potential problem for street segments that are too long, but still intersect with the buffer region around slums. Since properties farther from slums tend to have higher values, this could overestimate $SlumValue_{st}$.

To avoid this, I calculate the proportion of each segment that lies inside the buffer region and use these shares as weights when calculating $SlumValue_{st}$. This decreases the influence of segments that are predominantly outside the pre-defined radius. Formally, proxies for slum land value using data within distance r of slum s in year t are given by

$$SlumValue^{r}_{st} = \frac{\sum_{i=1}^{I} sh_length(i,s,r) * TotalNumerator_{it}}{\sum_{i=1}^{I} sh_length(i,s,r) * TotalDenominator_{it}} \tag{C.1}$$

Such that $TotalNumerator_{it}$ and $TotalDenominator_{it}$ are, respectively, the total built area and the total plot area in each street segment in the case of FAR. For Assessed Values, I divide total floor space price by total built area.

Moreover, $sh_length(i,s,r)$ is the length of segment i that lies inside the buffer region with radius r around slum s divided by its total length:

$$sh_length(i,s,r) = \frac{length(i,s,r)}{length_i} \tag{C.2}$$

Fig. C.4 illustrates this procedure with one slum (dark grey), a 500-meter buffer area around it (light grey), and street segments crossing this buffer. Average segment values are in colors.

Table C.1
Average price of new buildings and floor-to-area ratio (F.A.R.) across 96 districts in São Paulo, from 2001 to 2013.

	Price ('000 BRL per m^2)												
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
Constant	1.5***	1.5***	1.5***	1.4***	1.8***	1.8***	1.8***	1.8***	1.8***	2.1***	2.6***	2.8***	2.8***
	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)	(0.2)	(0.2)	(0.1)	(0.1)	(0.2)	(0.2)	(0.2)	(0.2)
F.A.R.	0.7***	0.6***	0.5***	0.7***	0.2**	0.2*	0.3***	0.3***	0.7***	0.7***	0.9***	0.5***	0.8***
	(0.1)	(0.1)	(0.09)	(0.1)	(0.09)	(0.1)	(0.1)	(0.09)	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)
Observations	73	71	73	73	69	66	75	78	75	72	76	79	77
R^2	0.34	0.37	0.27	0.36	0.09	0.05	0.11	0.11	0.35	0.31	0.41	0.16	0.43
Adjusted R^2	0.33	0.36	0.26	0.35	0.08	0.03	0.09	0.10	0.34	0.30	0.40	0.15	0.42

Notes: Dependent variable is the average price per square meter (in thousands of 2013 BRL) of new residential and commercial buildings in each district of São Paulo. Explanatory variable is the average floor-to-area ratio of all buildings in each district. Model is estimated for each year separately, as indicated by each column header. The city of São Paulo is divided into 96 districts. Data for new buildings is only available until 2013. *p<.1; **p<.05; ***p<.01.

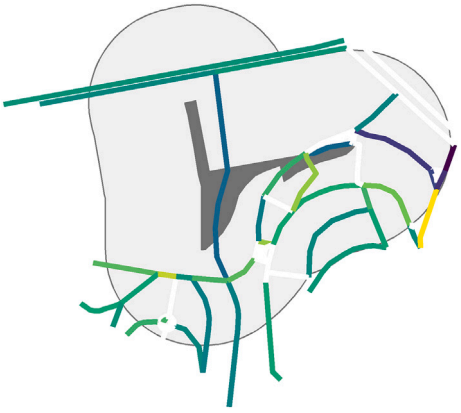


Fig. C.4. Example of street segments around slum with 500-meter buffer in 2011.
Notes: This figure shows a slum (dark grey) and a buffer area with a 500-meter radius (light grey). Street segments are colored from lower (darker colors) to higher (lighter colors) proxy values. Street segments in blank (white) had no properties found in tax data.

C.5. Slum and neighborhood characteristics

Fig. C.5 shows the distribution of the main covariates across FAR quantiles. Because census tracts and slums do not overlap perfectly, the covariates presented here represent a combination of slum and surrounding area characteristics. This is why the share of public services may seem high, for instance. Interestingly, the variable "Residents per Hectare" does not vary substantially across FAR quantiles because slums tend to occupy the space very efficiently no matter their location. Hence, density is quite homogeneous.

C.6. Slum fires and sorting

One alternative explanation is that fires could be driven by the sorting of either slum residents or worse infrastructure into neighborhoods where land value increased.

However, this does not seem to be the case in this context. Fig. C.6 shows correlations between changes in slum density or infrastructure and changes in land value, measured by FAR. The figure shows scaled differences between variables observed in 2010 and 2000. To calculate these differences, I use Census data, which cover a fraction of the slums

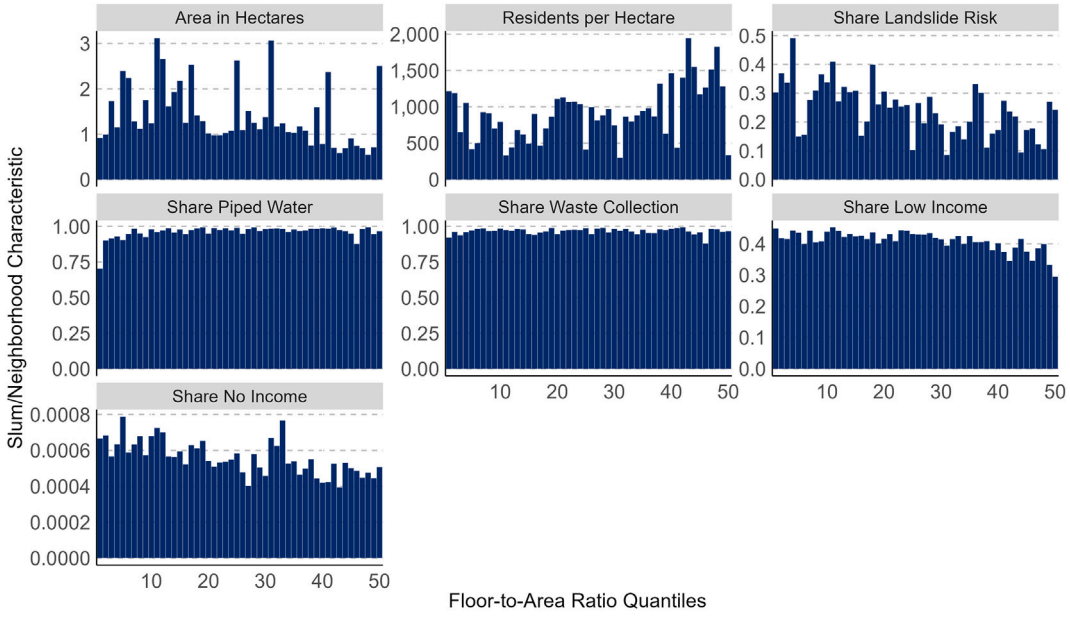


Fig. C.5. Slum and neighborhood characteristics and quantiles of floor-to-area ratio in 2001.
Notes: Each figure shows the distribution of slum and slum neighborhood characteristics across quantiles of Floor-to-Area Ratio. Slum characteristics include area and share of slum territory under landslide risk. Slum neighborhood characteristics are derived from the 2000 Census and include residents per hectare, coverage of piped water and waste collection, share of heads of household with no income, and share of household heads earning up to 3 times the Brazilian minimum wage (low income).

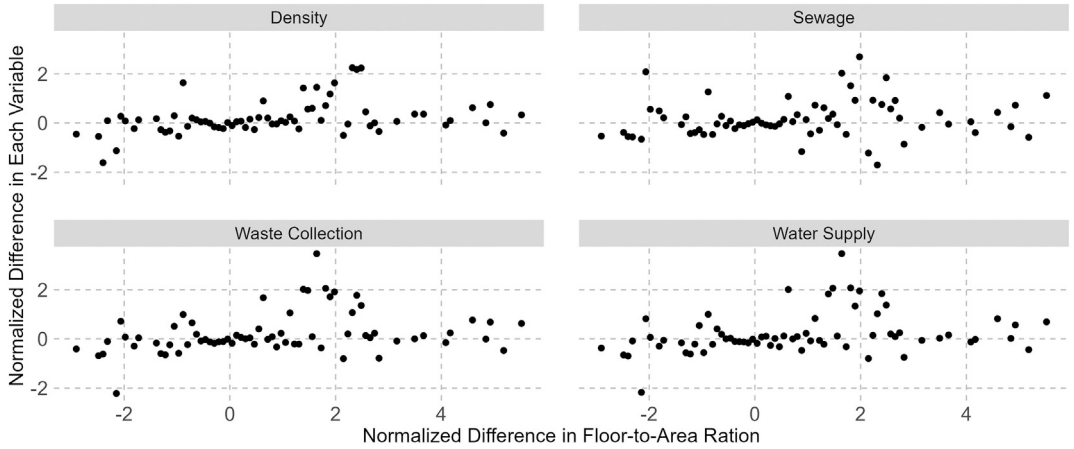


Fig. C.6. Increase slum density and infrastructure and changes in neighboring floor-to-area ratio between 2000 and 2010.
Notes: Figures show the correlation between the scaled (normalized) change in FAR and the scaled change in the variables indicated by each chart title. Scaled differences are calculated by computing the level of each variable in 2010 minus its level in 2000 and then normalizing the result. All variables are from Census data, which do not map all slums in the main sample. Quantities in this figure represent 993 slums that existed in both 2000 and 2010.

in my sample, but contain more information about number of residents and public service coverage. Moreover, I focus on slums that existed in both 2000 and 2010.[27]

These indicators do not seem to have a relevant correlation with land value. This suggests that something else is behind the non-linear pattern of slum fires across land value quantiles.

Appendix D. Additional results and robustness checks

D.1. Panel regression

D.1.1. Excluding unknown land ownership

In the main specification, the "public" category includes slums on land with unknown ownership. Fig. D.1 shows results when the latter are excluded from the sample.

D.1.2. Assessed value per square meter

Fig. D.2 presents results for a panel regression using quantiles of assessed value per square meter instead of FAR. I use values from the previous year to reduce potential simultaneity, since a slum removal could affect prices faster than FAR. The pointwise estimate for private slums in the top quantile is larger than in the main result, but the pattern is roughly the same.

D.1.3. Effect on covariates

Fig. D.3 shows the estimated differences in covariate levels between slums on private and public lands and across FAR quantiles. There is no evidence that infrastructure or density is significantly different on private lands compared with public lands in the highest quantiles.

D.1.4. Discontinuity in the probability of fire

To search for a discontinuity in the probability of fire, I divide the land value proxy into one hundred quantiles, which serve as cutoffs. For each cutoff, I construct a subsample that includes slums from up to ten quantiles above and up to ten quantiles below the cutoff, resulting in 15–20 quantiles per subsample. I do this separately for public and private slums and I compare the probability of fire for slums above and below each cutoff.

[27] I refrain from using this data to identify slum creation, because technological improvements allowed for more slums to be mapped in 2010 than in 2000.

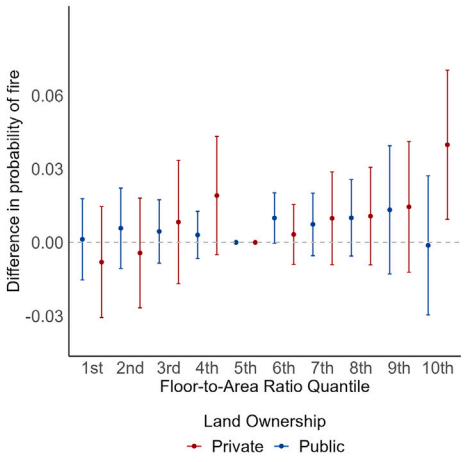


Fig. D.1. Estimated difference in probability of fire across floor-to-area ratio quantiles with full set of controls, 2001–2016 - excluding unknown land ownership.
Notes: Figure shows results for a linear probability model with the occurrence of slum fires on the left-hand side and ten quantiles of Floor-to-Area Ratio on the right-hand side. The specification includes district-year fixed effects as well as the following covariates interacted with year fixed effects: ten quantiles of area in square meters, ten quantiles of FAR prior to 2001, ten quantiles of number of properties per hectare, ten quantiles of residents per household, share of piped water, share of waste collection, share of residents with no income, share of residents with low income, and share of slum area under landslide risk. Sample excludes slums on land whose ownership is unknown. Errors in both specifications are spatially robust, based on (Conley, 1999).

I estimate Eq. (D.1) for each subsample around cutoff k, such that k ranges from 6 to 96. I do not run this for all percentiles to ensure a minimum of five quantiles to either side of the cutoff. Additionally, because samples are smaller, I use a parsimonious specification including only year fixed effects (τ).

$$ProbFire_{st} = \beta_1 \mathbb{1}[QuantileSV_{s,t} \geq k] + \tau_t + \epsilon_{st} \quad \text{(D.1)}$$

Subsamples exhibiting a discontinuity in the probability of fire should yield a significant estimate for β_1. Conversely, if the distribution

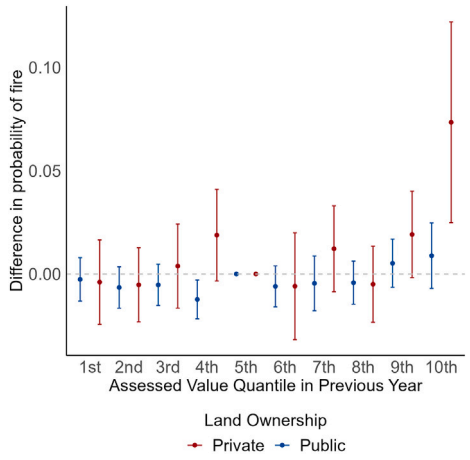


Fig. D.2. Estimated difference in probability of fire across assessed value quantiles with full set of controls, 2001–2016.
Notes: Figure shows results for a linear probability model with the occurrence of slum fires on the left-hand side and ten quantiles of Floor-to-Area Ratio on the right-hand side. The specification includes district-year fixed effects as well as the following covariates (measured before sample period) interacted with year fixed effects: ten quantiles of area in square meters, ten quantiles of FAR, ten quantiles of households per hectare, ten quantiles of residents per household, share of piped water and waste collection, share of residents with no income, share of residents with low income, and share of slum area facing landslide risk. Errors in both specifications are spatially clustered as in (Conley, 1999), with a 500-meter radius.

is smooth across land values, the estimated difference between slums above and below each cutoff should be close to zero. Fig. D.4 presents the estimated coefficient β_1 for each quantile cutoff.

For slums on private land, the estimated coefficient is significantly different from zero only around the 90th percentile cutoff. In contrast, for slums on public land, the estimates consistently remain close to zero. Moreover, coefficients for private slums are different from those for public slums only around the 90th percentile cutoff as well. These findings suggest a discontinuity around the 90th quantile only for slums on private land, aligning with the findings shown in Fig. 3(b).

D.2. Difference-in-differences

D.2.1. Descriptive figures for Água espraiada intervention

Fig. D.5 presents how the local housing market reacted to the urban intervention. Fig. D.5(a) shows requests for building permits. The y-axis shows the ratio between the stock of requests inside *Água Espraiada* and the stock outside the intervention, but within 4 km. This illustrates how fast the demand for building permits increased in a short period. In 2003, the stock of permit requests inside *Água Espraiada* was 50 % lower compared to neighboring areas. In 2007, requests inside *Água Espraiada* were almost twice as large as those outside the intervention.

This rapid growth in demand led to a delayed and slower increase in urban density. Fig. D.5(b) shows that the FAR inside *Água Espraiada* went from 10 % below to 5 % above the FAR outside the intervention zone. Combined, both panels in Fig. D.5 suggest that land value inside *Água Espraiada* increased after 2004 compared to neighboring areas.

D.2.2. Inverse probability weighting procedure (IPW)

In this section, I detail the IPW procedure used in the Difference-in-Differences estimation. First I run a probit model in Eq. (D.2) to estimate the probability of treatment $\hat{p}$ as a function of observable covariates.

$$Prob(Treated_i = 1|X_i) = \Phi(X_i\alpha) \tag{D.2}$$

such that $\Phi(\cdot)$ is a standard cumulative Normal distribution and α is a vector of coefficients estimated by maximum likelihood.

Then, I use the estimated propensity scores $\hat{p}$ to re-weigh the main regression model from Eq. (3). I attribute weights $1/\hat{p}$ to treated units and $1/(1-\hat{p})$ to control ones. Moreover, I remove any observation with $0.05 \leq \hat{p} \leq 0.95$ to avoid over-weighing some observations (Cunningham, 2021).

D.2.3. Propensity score estimation

Tables D.1 and D.2 present the estimated propensity score models for private and public slum-occupied land, respectively, based on a maximum distance threshold of 4 km.

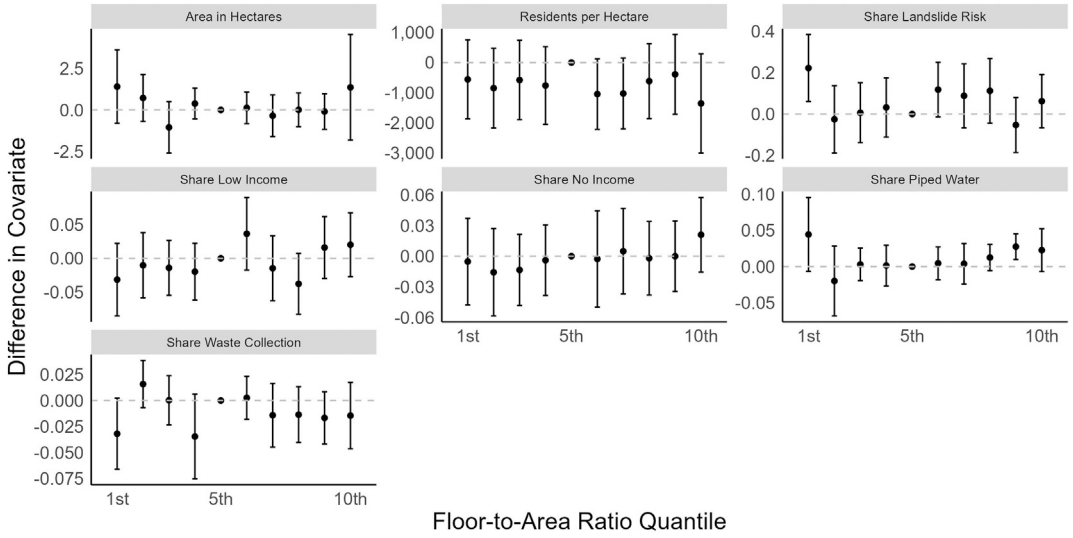


Fig. D.3. Regression of main covariates on quantiles for slums on private and public lands.
Notes: Figure shows the difference in covariates between slums on private and public lands. The empirical model is similar to the one in Fig. 3(b) for each selected covariate, except that observations are restricted to Census Year 2000 and that the estimated coefficients represent the difference between private and public lands. All specifications include district fixed effects as well as the following covariates: ten quantiles of area in square meters, ten quantiles of FAR prior to 2001, ten quantiles of residents per household, share of piped water, share of trash collection, share of residents with no income, share of residents with low income, and share of slum area under landslide risk. Errors are spatially robust, based on (Conley, 1999).

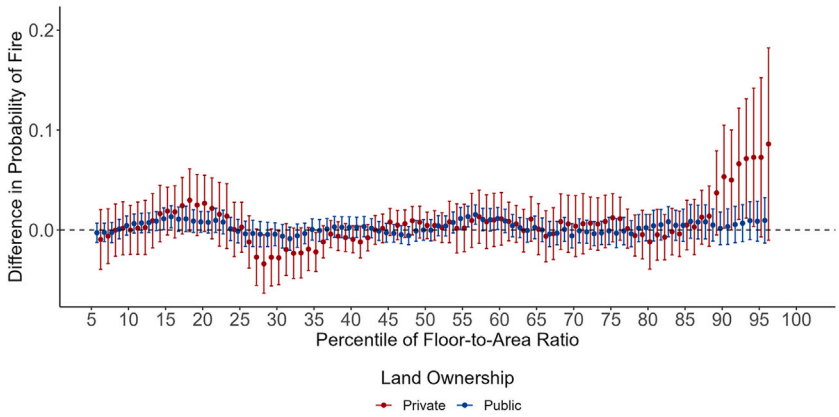
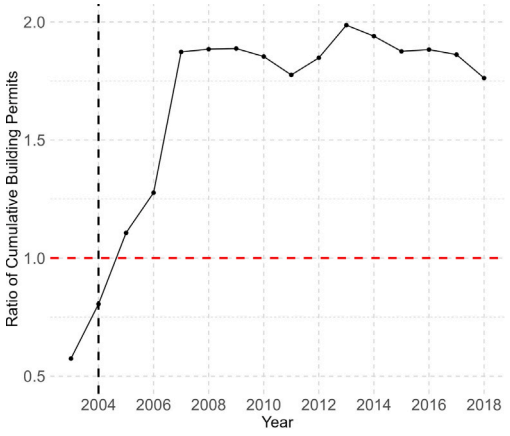


Fig. D.4. Rolling window difference in probability of fire for slums on private and public lands.

Notes: Figure shows the estimated difference in probability of fire between high and low land values for multiple cutoffs in the FAR distribution. Each estimate is based on a rolling-window sub-sample with 5–10 percentiles on each side of the kth percentile. For example, if $k = 15$, the sample includes only percentiles ranging from 5th to 24th. In this case, the estimate compares the average probability of fire in percentiles 15 to 24 versus percentiles 5 to 14. For $k = 96$, quantiles 96 to 100 are above the cutoff and quantiles 86 to 95 are below it. The same reasoning applies to all k percentiles such that $k \in \{6, 7, \ldots, 96\}$. All specifications include year fixed effects with spatially clustered errors as in Conley (1999).



(a) Ratio of Stocks of Building Permit Requests Inside and Within 4 Kilometers of Urban Intervention Água Espraiada



(b) Ratio of Floor-to-Area Ratios Inside and Within 4 Kilometers of Urban Intervention Água Espraiada

Fig. D.5. Building permits, floor-to-area ratio, and urban intervention Água Espraiada.

Notes: Panel (a) shows the ratio between the stock of building permit requests inside *Água Espraiada* and the stock outside the intervention, but within 4 km. Permit data is only available starting in 2003. Panel (b) shows the ratio of FAR inside *Água Espraiada* and FAR outside the intervention, but within 4 km.

Table D.1
Probit regression of probability that slum is inside urban intervention on pre-treatment characteristics - private slums within 4 km.

Share water	7.40 (9.26)
Share waste collection	−4.17 (7.02)
Share income up to 3 salarios minimos	2.99 (1.88)
Share of no income	−0.09 (2.20)
Residents per household	0.61 (0.79)
Households per hectare	−0.0001 (0.0001)
(log) Neighborhood F.A.R.	2.66*** (0.61)
(log) Slum area	0.03 (0.23)
Share landslide risk	−5.23** (2.56)
Constant	−5.87 (6.95)
Pseudo-R2	0.41
Observations	165

Notes: Slums inside and within 4 km of Urban Intervention's border. *p<.1; **p<.05; ***p<.01

Table D.2
Probit regression of probability that slum is inside urban intervention on pre-treatment characteristics - public slums within 4 km.

Share water	−1.80 (1.70)
Share waste collection	−6.17*** (1.90)
Share income up to 3 salarios minimos	1.72** (0.87)
Share of no income	−1.91 (1.22)
Residents per household	1.49*** (0.42)
Households per hectare	−0.0001 (0.0001)
(log) Neighborhood F.A.R.	3.13*** (0.36)
(log) Slum area	−0.39*** (0.12)
Share landslide risk	−1.03** (0.47)
Constant	2.02 (2.15)
Pseudo-R2	0.4
Observations	329

Notes: Slums inside and within 4 km of Urban Intervention's border. *p<.1; **p<.05; ***p<.01

The following variables are from the 2000 Census: *Share Water*, *Share Waste Collection*, *Share Income up to 3 SalÁrios Mínimos*, *Share No Income*, *Residents per Household*, and *Households per Hectare*. The *(log) Neighborhood F.A.R.* is from property tax data in 1996. *Slum Area* and *Share Landslide Risk* are geographical features from municipal government data for slums that existed in 2000.

Figs. D.6(a) and D.6(b) provide evidence of common support for treated and control units among private and public slums respectively.

Balance for slums on private land is provided in main text. Table D.3 shows balance in observables for slums on public land.

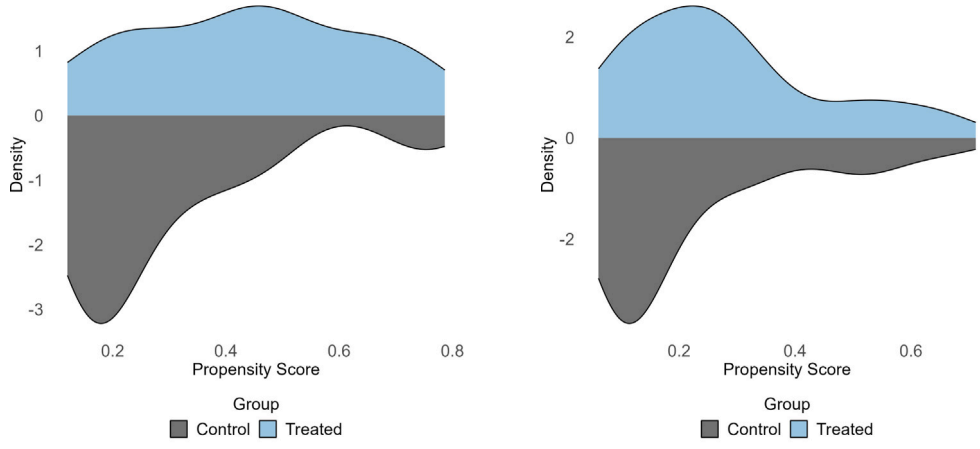
D.2.4. Unmatched sample

Table D.4 presents results for the unmatched Difference-in-Differences (without using IPW). Results are similar to the re-weighed model once I control for all covariates and fixed effects. This suggests that the inclusion of observable characteristics, either explicitly or via IPW, helps smoothing potential pre-treatment differences across treated and control groups.

Fig. D.7 shows that results are stable and similar to the ones from the matched sample. However, their significance depends more on the maximum distance threshold in the absence of IPW. This is likely attributable to the fact that slums farther apart might be, on average, less comparable to the treated group.

D.2.5. Restricting the sample period

In Table D.5, I re-estimate the main specification, but now restricting the sample to different periods. For each column, the header indicates first and last years of sample. For example, in Column (1), I estimate the model for years ranging from 2001 to 2016 (full sample). In Column (2), the sample spans from 2001 to 2015. In Column (3), it covers from 2001 to 2014. And so on and so forth.



(a) Slums on Private Land (b) Slums on Public Land

Fig. D.6. Distribution of propensity scores for treated and control slums within 4 km of Agua Espraiada.
Notes: Treated units are all slums inside the perimeter of the urban intervention. Control units are all slums located both farther than 500 m and within 4 km of the urban intervention. Units with estimated propensity scores higher than 0.95 or lower than 0.05 were discarded.

Table D.3
Propensity-score matching balance test for slums in public land inside or within 4 km of urban intervention.

	Unmatched			Matched		
	Treated (1)	Control (2)	Difference (3)	Treated (4)	Control (5)	Difference (6)
(log) Neighborhood F.A.R.	−0.324	−0.757	0.433 (0.059)	−0.437	−0.397	−0.039 (0.051)
Households per hectare	1783.809	1988.811	−205.002 (929.946)	1832.524	1793.859	38.665 (403.166)
(log) Slum area	−1.466	−1.014	−0.452 (0.248)	−1.372	−1.486	0.113 (0.226)
Residents per household	3.738	3.701	0.037 (0.049)	3.709	3.693	0.016 (0.056)
Share income up to 3 salarios minimos	0.379	0.389	−0.01 (0.022)	0.355	0.348	0.007 (0.025)
share of no income	0.132	0.148	−0.016 (0.016)	0.142	0.127	0.015 (0.016)
share landslide risk	0.048	0.219	−0.171 (0.054)	0.127	0.054	0.072 (0.042)
share waste collection	0.964	0.981	−0.017 (0.008)	0.971	0.966	0.005 (0.012)
Share water	0.918	0.983	−0.065 (0.013)	0.973	0.97	0.003 (0.012)
Observations	34	295	-	25	76	-

Notes: Columns (1) and (2) show the means of each variable conditional on treatment status. Columns (4) and (5) show the re-weighed means such that treated and control slums receive weights 1/p and 1/(1-p) respectively. Columns (3) shows the estimated coefficient of an OLS regression of each variable on treatment. Columns (6) repeats the regression in Column (3), but re-weighing according to (4) and (5).

Table D.4
Effect of urban intervention on strategic arson in slums on private land.

Dependent variable:	Probability of fire			
Model:	(1)	(2)	(3)	(4)
Inside OUC	0.02 (0.04)			
Year ≥ 2005	−0.002 (0.006)	−0.002 (0.006)		
Inside OUC × Year ≥ 2005	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.06** (0.03)
Slum FE (165)		Yes	Yes	Yes
Year FE (15)			Yes	Yes
Covariates * year				Yes
Observations	2475	2475	2475	2475
R^2	0.007	0.293	0.296	0.337
Within R^2		0.001	0.001	0.059

Notes: Slums inside or within 4 km of Urban Intervention. Columns (1)–(4) report effects on probability of fire. Column (1) has no controls. Column (2) adds slum fixed effects; Column (3) adds year fixed effects; Column (4) adds interaction between covariates and year fixed effects. Spatially robust errors are based on Conley (1999) with 500-meter distance threshold. *p<.1; **p<.05; ***p<.01.

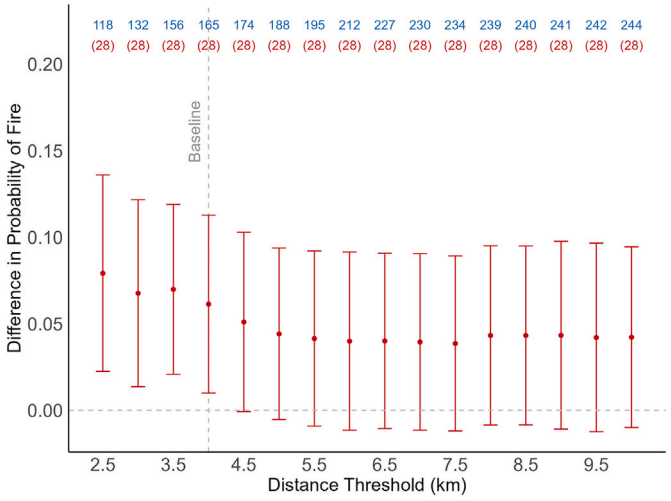


Fig. D.7. Effect on probability of fire for each maximum distance threshold - unmatched sample of private slums.
Notes: The total number of observations for each threshold is in blue, whereas the number of treated units is in red and in brackets. Treated units are all slums inside the perimeter of the urban intervention. Control units are all slums located farther than 500 m from the urban intervention and within each distance threshold.

Table D.5
Effect of urban intervention on probability of fire in slums on private land - alternative sample periods.

Dependent variable:	Probability of fire									
Sample period:	2001–2016	2001–2015	2001–2014	2001–2013	2001–2012	2001–2011	2001–2010	2001–2009	2001–2008	2001–2006
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Inside OUC × Year ≥ 2005	0.07***	0.07***	0.08***	0.07***	0.07***	0.07***	0.06***	0.08***	0.08***	0.05**
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.02)
Slum FE (56)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
IPW covariates * Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# Year FE	15	14	13	12	11	10	9	8	7	6
Observations	840	784	728	672	616	560	504	448	392	336
R^2	0.523	0.541	0.553	0.560	0.555	0.580	0.618	0.627	0.625	0.684
Within R^2	0.248	0.255	0.268	0.278	0.284	0.301	0.285	0.282	0.272	0.322

Notes: Slums located inside or within 4 km of Urban Intervention. Inverse Propensity Score Weighing (IPW) applied to all specifications such that treated units receive 1/p weights whereas control ones receive 1/(1-p) weights. Each column header indicates the sample period. All columns include slum and year fixed effects, as well as interaction between IPW covariates and year fixed effects. Spatially robust errors are based on Conley (1999) with 500-meter distance threshold. *p<.1; **p<.05; ***p<.01.

I also test the exclusion of pair-wise combinations of years. Each estimate in Fig. D.8 corresponds to the main Difference-in-Differences specification for private slums based on a sample that includes all but two years from the 2001–2016 period. Each time, a different pair is dropped from the sample. Results are generally robust, indicating that heterogeneous geocoding success rates are not driving the main results.

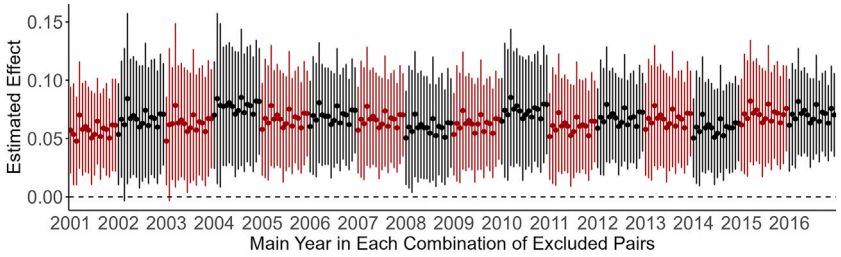


Fig. D.8. Effect of urban intervention on strategic arson in slums occupying private lands - combinatorial exclusion of years in sample.
Notes: Each estimate is based on a sample that excludes pair-wise combinations of years ranging from 2001 to 2016 (with repetition). The years on the x-axis represent the main element of the pair, which is combined with all years (2001–2016) sequentially. Alternating colors indicate when the main year in the pair changes. For example, the first chunk of estimates in red is based on samples that exclude combinations of 2001 with all years (2001–2016). The second chunk, in black, is based on samples that exclude combinations of 2002 with all years (2001–2016). Displaying 95 % Confidence Intervals with Spatially Robust Errors as in Conley (1999).

D.2.6. Excluding unknown land ownership

Table D.6
Effect of urban intervention on strategic arson in slums occupying public lands - excluding unknown land ownership.

Dependent variable:	Probability of fire			
Model:	(1)	(2)	(3)	(4)
Inside OUC	−0.005 (0.02)			
Year ≥ 2005	0.006 (0.01)	0.006 (0.01)		
Inside OUC × year ≥ 2005	−0.009 (0.01)	−0.009 (0.01)	−0.009 (0.01)	−0.02* (0.01)
Slum FE (59)		Yes	Yes	Yes
Year FE (15)			Yes	Yes
IPW covariates * year				Yes
Observations	885	885	885	885
R^2	0.002	0.140	0.155	0.402
Within R^2		0.000	0.000	0.292

Notes: Slums inside or within 2006 km of Urban Intervention. Inverse Propensity Score Weighing (IPW) applied to all specifications such that treated units receive 1/p weights whereas control ones receive 1/(1-p) weights. Column (1) has no controls. Column (2) adds slum fixed effects; Column (3) adds year fixed effects; Column (4) adds interactions between IPW covariates and year fixed effects. Spatially robust errors are based on Conley (1999) with 500-meter distance threshold. *p<.1; **p<.05; ***p<.01.

Data availability

Data will be made available on request.

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